

Self-oscillation in a thermoresponsive gel as a relaxation cycle on a spatially generated slow manifold

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A thermoresponsive hydrogel slab driven by an exothermic reaction sustains volume oscillations whose dominant mode, we show, is a relaxation oscillator on a slow manifold that is generated by spatial structure and has no counterpart in any spatially uniform reduction. We formulate the coupled thermo-chemo-poroelastic model, derive the dispersion relation about every uniform branch, and trace the parameter plane (Bi_T, S_χ) from a 25×25 default-IC PDE sweep. The realized attractor across $\sim 45\%$ of the plane is a four-phase cycle in which the surface trajectory traverses the full S-curve of $\mu(J, \theta) = \mu_b$ between two folds while the bulk sits buffered on the swollen branch; the slow phase is set by the φ -dependent diffusion barrier. The same barrier suppresses runaway global collapse, fixing the LCST-front penetration depth at $\xi_{LCST} \simeq 0.885$ at the working point. A closed-form inversion of the homogeneous steady-state surface shows that this cycle persists over a $\sim 51\%$ “collapsed-only” subregion in which no swollen homogeneous branch exists for a 0D linearisation to sit on, and the upper- S_χ exit obeys SNIC scaling $T \propto (S_\chi^c - S_\chi)^{-1/2}$ rather than a logarithmic homoclinic law—a global topological event in the slow-manifold plane, not a local linear bifurcation. Initial-condition sweeps in a representative bistable cell reveal a hidden hot-runaway steady state coexisting with the cycle, producing a sharp $\langle J \rangle$ jump at $\theta_0^* \approx 2.4$ that is a falsifiable prediction beyond the spatially uniform CSTR analogue. The findings establish linear stability as necessary but not sufficient for this class of gel oscillator and identify the spatially generated slow manifold as the organising structure.

I. INTRODUCTION

Stimuli-responsive gels that undergo spontaneous, self-sustained volume oscillations have attracted attention as minimal models of biological rhythms and as candidate components for soft actuators, chemical oscillators, and autonomous drug-delivery systems [1, 4, 13]. The most studied synthetic realization couples a thermoresponsive network—typically based on poly(*N*-isopropylacrylamide) (PNIPAM) with a lower critical solution temperature (LCST) around 33°C [11, 14]—to an oscillatory or exothermic chemical reaction such as the Belousov–Zhabotinsky (BZ) reaction [2, 3]. In these systems, chemical oxidants produced by the BZ reaction swell the gel, while reductants cause collapse; the gel’s spatiotemporal dynamics feed back on the local oxidant concentration, giving rise to autonomous oscillation [4].

A complementary oscillation route, less well studied, relies on purely *thermal* feedback: an exothermic reaction heats the gel above the LCST, the attendant volume collapse expels solvent and reactant, the gel cools, swells again, and re-ingests fresh reactant—completing one cycle. This thermo-mechanical-chemical feedback loop does not require multi-step BZ chemistry; a single irreversible exothermic reaction suffices. Early theoretical work by Siegel and coworkers demonstrated this concept for spherical gel beads [5, 6], and subsequent studies identified parameter regimes of self-oscillation in slab and membrane geometries [7–9]. The underlying thermal-runaway instability has classical antecedents in combustion theory [16] and exothermic CSTR dynamics [18], but its coupling to gel poroelasticity introduces the transport-suppression negative feedback that enables

sustained oscillation rather than single ignition events. Two questions remain open even for the simplest planar slab geometry, which we adopt throughout this paper:

1. What is the spatially resolved mechanism that organises the thermal-LCST cycle, and is it predicted by the linear stability of a spatially uniform branch?
2. Does the spatial coupling enrich the attractor inventory beyond the cycle—and if so, can the additional attractors coexist with it under identical bath conditions?

Section II fixes the thermo-chemo-poroelastic model in the planar slab and reduces the parameter space to eight dimensionless groups. Section III derives the dispersion relation, identifies the long-wave Hopf window, and separates the oscillatory mode from the finite-wavenumber Cahn–Hilliard spinodal mode. Section IV turns to the full PDE in four steps: it reports the realized attractors of the default-IC sweep (Sec. IV A), shows that the dominant attractor is a relaxation oscillator on a spatially generated slow manifold (Sec. IV B), proves that this slow manifold exists over a parameter region where no homogeneous swollen branch is available for the 0D analysis to linearise about (Sec. IV D), and uncovers a hidden hot-runaway steady state that produces a falsifiable IC-induced hysteresis prediction (Sec. IV E). Full derivations and numerical details appear in Appendices A–D.

II. MODEL

A. Physical system

We consider a planar thermoresponsive gel slab of reference half-thickness H_0 containing an immobilised catalyst that catalyses an exothermic, irreversible, first-order reaction. The slab is in contact at its free surface with a well-mixed bath of constant temperature T_∞ and constant reactant concentration c_b ; $\xi = X/H_0 \in [0, 1]$ is a Lagrangian coordinate, with $\xi = 0$ the symmetry plane and $\xi = 1$ the free surface. The spherical and membrane variants of the same reaction–swelling coupling [5, 6, 8, 9] introduce additional radial-stretch and curvature effects that we defer to a separate study.

B. Governing equations

The three primary fields are the local swelling ratio J , the normalized reactant concentration $u = c/c_b$, and the dimensionless temperature excess $\theta = (T - T_\infty)/\Delta T_*$, where $\Delta T_* = (-\Delta H_r)_{c_b}/C_*$ is the adiabatic reaction temperature rise. The dimensionless governing equations (derived in Appendix B) are

$$\frac{\partial J}{\partial t} = \frac{\partial}{\partial \xi} \left[\mathcal{M}(J) \frac{\partial \mu}{\partial \xi} \right], \quad (1a)$$

$$\frac{\partial (uJ)}{\partial t} = -\frac{\partial}{\partial \xi} (uq) + \frac{\partial}{\partial \xi} \left[\mathcal{D}(J) \frac{\partial u}{\partial \xi} \right] - \text{Da} J \hat{R}, \quad (1b)$$

$$\mathcal{C}(J) \frac{\partial \theta}{\partial t} = \frac{\partial}{\partial \xi} \left[\mathcal{K}(J) \frac{\partial \theta}{\partial \xi} \right] + \text{Da} J \hat{R}, \quad (1c)$$

where $q = -\mathcal{M} \partial_\xi \mu$ is the solvent flux, $\mathcal{M}$ is the swelling mobility, $\mathcal{D}$ and $\mathcal{K}$ are the effective reactant diffusivity and thermal conductivity (with the prefactors δ and α absorbed; see Appendix B), and $\mathcal{C}$ is the effective heat capacity. The enthalpy-advection term ($\text{Pe}_T \ll 1$) is neglected. The reaction rate is

$$\hat{R} = u (1 - \phi)^{m_{\text{act}}} \exp\left(\frac{\Gamma_A \theta}{1 + \varepsilon_T \theta}\right), \quad (2)$$

with $\phi = \phi_{p0}/J$. The solvent chemical potential $\mu[J, \theta]$ is derived from the Flory–Rehner free energy with a temperature-dependent χ parameter exhibiting an LCST transition (Appendix A).

C. Boundary conditions

At the symmetry plane ($\xi = 0$) all fluxes vanish. At the free surface ($\xi = 1$):

$$q|_1 = \text{Bi}_\mu \mu, \quad \mathcal{D} u_\xi|_1 = \text{Bi}_c (1 - u), \quad \mathcal{K} \theta_\xi|_1 = -\text{Bi}_T \theta, \quad (3)$$

where the bath chemical potential is set to $\mu_b = 0$ by convention.

D. Dimensionless parameters

The dimensionless groups are summarized in Table I.

TABLE I. Dimensionless parameters: primary control parameters (top) and secondary material/numerical parameters (bottom).

Symbol	Physical meaning	Default	Range
Da	Reaction rate / swelling rate	4	3–18
Bi_μ	Surface permeation Biot number	1.0	—
Bi_c	Surface reactant Biot number	0.70	0.1–2
Bi_T	Thermal exchange number	0.10	0.05–0.5
S_χ	LCST sensitivity	1.0	0.5–8
Γ_A	Arrhenius sensitivity	1.5	0–7
D_0	Reactant diffusivity scale	2.0	0.5–3
m_{act}	Reaction accessibility exponent	6	1–6
m_{diff}	Diffusion tortuosity exponent	2	1–6
α	Thermal / swelling time ratio	0.03	—
δ	Diffusion / permeation ratio	0.08	—
ℓ	Interface regularization length	0.01	—
Ω_e	Elastic / mixing energy ratio	0.12	—
ϕ_{p0}	Reference polymer fraction	0.15	—
χ_∞	Interaction parameter at T_∞	0.60	—
χ_1	Concentration-dependent χ	1.10	—
ε_T	Nonlinear Arrhenius correction	0.03	—

III. LINEAR STABILITY ANALYSIS

A. Dispersion relation and base state

The stability of spatially uniform states of the PDE system (1) can be determined analytically. The uniform steady states (J_0, u_0, θ_0) satisfy three algebraic balances:

$$\begin{aligned} \mu(J_0, \theta_0) &= \mu_b, & \text{Bi}_c(1 - u_0) &= \text{Da} J_0 \hat{R}_0, \\ \text{Bi}_T \theta_0 &= \text{Da} J_0 \hat{R}_0, \end{aligned} \quad (4)$$

where $\hat{R}_0 = \hat{R}(u_0, J_0, \theta_0)$. Linearizing about this base state with normal-mode perturbations $\propto e^{\sigma t + i k \xi}$ yields a 3×3 eigenvalue problem for each slab wavenumber k (Appendix C):

$$\sigma \hat{\mathbf{y}} = \mathbf{A}(k) \hat{\mathbf{y}}, \quad \mathbf{A}(k) = \mathbf{A}_0 + k^2 \mathbf{D}_2 + k^4 \mathbf{D}_4. \quad (5)$$

Here $\mathbf{A}_0$ encodes reaction kinetics and surface exchange, $\mathbf{D}_2$ collects diffusive transport, and $\mathbf{D}_4$ arises from the Cahn–Hilliard gradient-energy regularization. Uniform branches are classified as *stable* ($\text{Re} \sigma_1(0) < 0$), *Hopf unstable* ($\text{Re} \sigma_1(0) > 0$, $\text{Im} \sigma_1(0) \neq 0$), or *monotone unstable* ($\text{Re} \sigma_1(0) > 0$, $\text{Im} \sigma_1(0) = 0$).

B. Two instability mechanisms

The dispersion relation reveals two physically distinct instabilities (Fig. 1).

a. Hopf instability ($k = 0$). The long-wave mode is governed by $\mathbf{A}_0$ alone. The positive feedback originates in the heat equation: when the Arrhenius heat–reaction coupling $\text{Da } J_0 R_\theta$ exceeds the surface heat loss Bi_T , a temperature perturbation amplifies the reaction rate, which in turn raises the temperature further. The negative feedback follows the off-diagonal loop $\theta \rightarrow J \rightarrow u \rightarrow \theta$: rising temperature increases χ and drives collapse, collapse suppresses reactant supply, and reduced reactant weakens the heat source. Oscillation requires these two feedback paths to operate on well-separated timescales; the onset condition is set by the balance of the characteristic-polynomial coefficients of $\mathbf{A}_0$ (Appendix C).

b. Spinodal instability ($k > 0$). When the thermodynamic derivative $f_J \equiv \partial_J[m_{\text{mix}} + m_{\text{el}}] < 0$, the base state lies inside the Flory–Huggins spinodal region and the $(1, 1)$ entry of $\mathbf{D}_2$ acts as a negative diffusivity: spatial perturbations in J are amplified rather than damped [19, 20]. The k^4 term limits growth to a finite band with maximum at $k^* \approx \sqrt{-f_J/(2\ell^2)}$ [Fig. 1(c)]. When triggered in isolation ($\text{Da} = 0$, fixed θ above LCST), this mode produces alternating swollen and collapsed domains (Appendix E).

c. Why spinodal does not manifest during oscillation. Although the spinodal growth rate ($\sigma^* \sim 10^3$) far exceeds the Hopf rate (~ 1), three factors prevent spinodal patterns from developing during the oscillation cycle: (i) the system traverses the spinodal region only transiently during collapse ($\lesssim 0.5 \tau$, as confirmed in Sec. IV), too briefly for short-wavelength domains to reach macroscopic amplitude; (ii) the Thiele penetration gradient and Robin boundary conditions break the translational symmetry required for classical spinodal pattern selection; (iii) the large-amplitude Hopf limit cycle continuously homogenizes the spatial structure at each period.

C. Implications for the working point

The lower- J (collapsed) branch at the default parameters undergoes a sequence monotone $\rightarrow$ Hopf $\rightarrow$ stable as Bi_T increases: weak cooling causes thermal runaway, intermediate cooling produces oscillatory growth, and strong cooling stabilizes the system [Fig. 1(a,b)]. The Hopf region persists only when Da and S_χ are large enough to overcome surface heat loss [Fig. 1(d,e)]. The linear period estimate $T_{\text{LSA}} = 2\pi/\omega$ [Fig. 1(f)] provides the small-amplitude precursor to the nonlinear oscillation period; the dispersion analysis correctly identifies the existence, type, and timescale of the instability that seeds the slab dynamics.

IV. RESULTS AND DISCUSSION

The dispersion analysis of Sec. III establishes the linear bifurcation conditions for the homogeneous swollen state. We now turn to direct simulations of the full PDE system (1) in the planar slab and report four results. First, default-IC simulations across the principal parameter planes realize three distinct attractors—cold-swollen, an LCST-front oscillation, and a frozen partial-collapse front (Sec. IV A). Second, the LCST-front attractor is a relaxation oscillator on a slow manifold generated by the surface boundary condition together with the φ -dependent diffusion barrier; the same barrier prevents the gel from undergoing global collapse (Sec. IV B). Third, the 0D linear stability analysis is a necessary but not sufficient predictor of the PDE oscillation: a closed-form parameter inversion of the homogeneous-SS conditions identifies a Whitney-cusp catastrophe whose collapsed-only branch covers $\sim 51\%$ of the displayed (Bi_T, S_χ) plane, and the saddle–node-on-invariant–circle scaling at the upper- S_χ boundary confirms that the cycle is destroyed by a global topological event rather than by a local Hopf instability of any homogeneous branch (Sec. IV D). Fourth, initial-condition sweeps reveal a fourth attractor—a hot-runaway super-swollen steady state—coexisting with the LCST-front cycle in a substantial portion of the C-region; the bistability gives a falsifiable IC-induced hysteresis prediction (Sec. IV E).

A. Realized attractors and their spatiotemporal structure

We integrate the PDE system from the canonical cold-swollen initial condition ($J_0 = 1.30$, $\theta_0 = 0$, $u_0 = 0.02$) to dimensionless time $t = 200$, classify each parameter point by combining the surface swelling amplitude, the peak polymer fraction, and the oscillation period into a six-class regime label, and sweep 25×25 points in the (Bi_T, S_χ) plane and 25×15 points in the (Bi_T, Da) plane (Fig. 2). Three of the six classification labels are populated:

- (i) a spatially uniform *cold-swollen* steady state on the swollen branch of the chemical-potential locus, occupying the cold edge of the phase diagram (low S_χ or high Bi_T); “cold” here refers to the Flory–Huggins branch and not to a literal bath temperature, since the chemistry continues to heat the gel even when no front nucleates;
- (ii) a large-amplitude *LCST-front* oscillation occupying the central $\sim 45\%$ of the (Bi_T, S_χ) plane, in which a narrow surface shell cycles between collapsed and swollen states while the centre stays cold-swollen;
- (iii) a *frozen partial-collapse* profile at large Bi_T , with

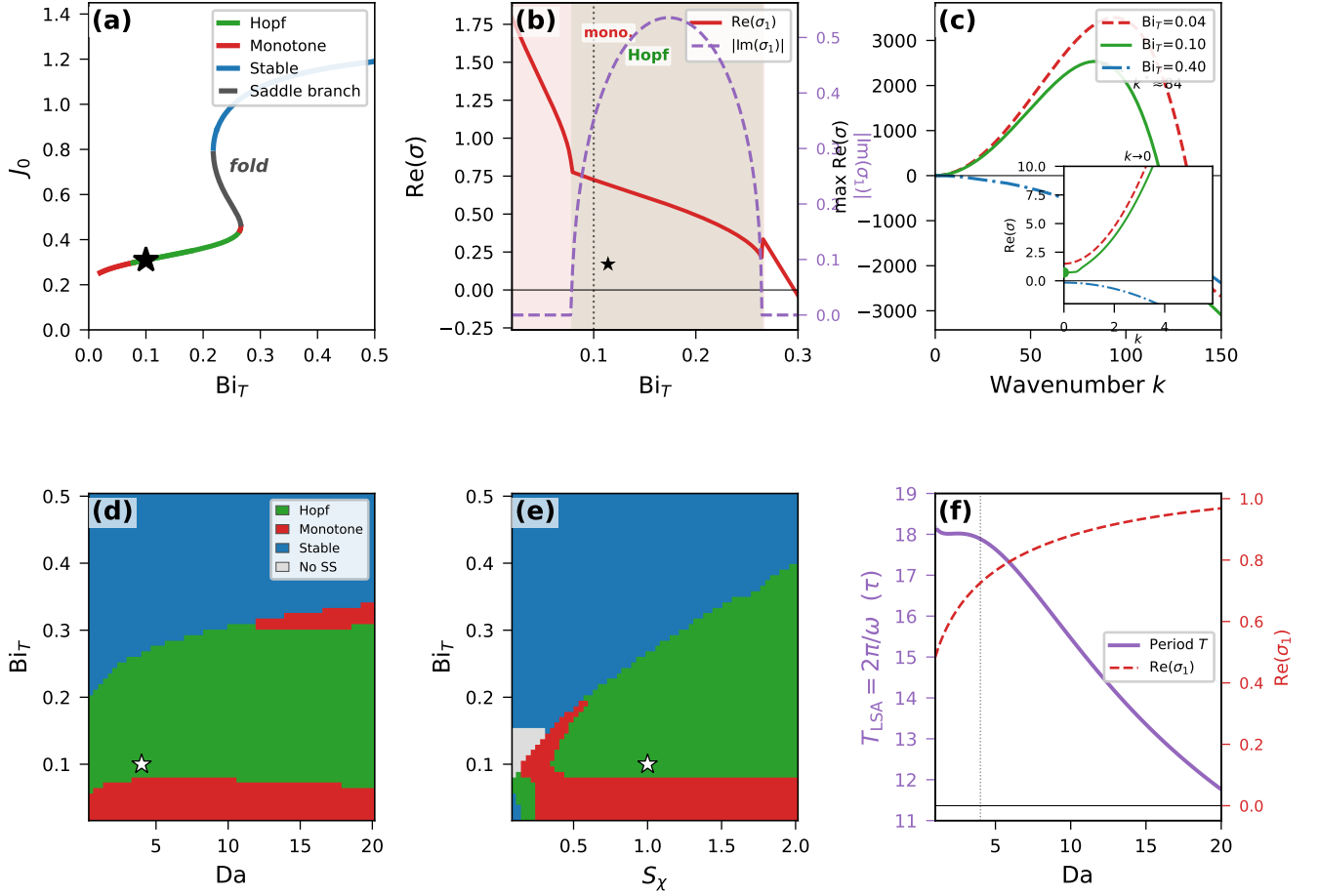


FIG. 1. Linear stability of the 1D slab. (a) Steady-state branches J_0 versus Bi_T , colored by $k = 0$ stability; the star marks the working point. (b) Leading eigenvalue (real and imaginary parts) along the collapsed branch, showing the monotone-to-Hopf transition. (c) Full dispersion curves for three Bi_T values; inset resolves the $k \rightarrow 0$ Hopf regime. (d,e) Stability maps in the (Da, Bi_T) and (S_χ, Bi_T) planes. (f) Linear period $T_{LSA} = 2\pi/\omega$ and growth rate vs Da .

the same two-zone spatial structure as the LCST-front oscillation but stationary in time.

The two remaining classifier-allowed regimes—globally collapsed steady and globally collapsing oscillation, both with $\varphi > 0.5$ throughout the gel at peak—never appear in the 1000 default-IC simulations of the two parameter sweeps. They are suppressed by two φ -dependent factors that act on the chemistry once a surface shell crosses the LCST and reaches the cycle’s peak collapse $\varphi \simeq 0.98$. The catalyst-accessibility factor $(1-\varphi)^{m_{act}}$ with $m_{act}=6$ multiplies the reaction rate and is suppressed by many orders of magnitude relative to its swollen-branch value, quenching the chemistry inside the collapsed shell. The reactant-transport factor $(1-\varphi)^{m_{diff}}$ in $D(\varphi)$, with $m_{diff}=2$, drops by $\sim 10^3$ over the same interval and isolates the deeper interior from any external reactant supply that might otherwise re-ignite collapse there. At the working point the two factors together fix the front penetration depth at $\xi_{LCST} \simeq 0.885$, leaving the inner 88.5% of the slab permanently below the LCST. Within the

present model this self-limiting structure is the principal reason a thermo-LCST gel does not undergo runaway global collapse despite the strongly exothermic Arrhenius kinetics [16, 18]. A fourth attractor, a uniform *hot-runaway* steady state, occupies part of the LCST-front parameter region but is inaccessible from the default cold IC; its analysis is deferred to Sec. IV E.

Figure 3 shows the spatiotemporal fields $J(\xi, t)$, $\theta(\xi, t)$ and the surface and centre $J(t)$ traces for one representative point per attractor.

The cold-swollen state [Fig. 3(a-c)] is spatially uniform but is not at the bath temperature: the Arrhenius source and the surface heat loss equilibrate at a homogeneous profile [$\theta \simeq 6.9$, $J \simeq 1.18$, $\varphi \simeq 0.13$ at the representative point]. The LCST coupling $S_\chi \theta \varphi^2$ is too weak at this S_χ to push χ_{eff} past the spinodal, so φ remains well below the LCST threshold and no front is nucleated.

The LCST-front cycle [Fig. 3(d-f)] has a two-zone structure. The outer shell ($\xi \in [0.885, 1]$ at the working point) cycles between the swollen state ($J \sim 1.4$) and

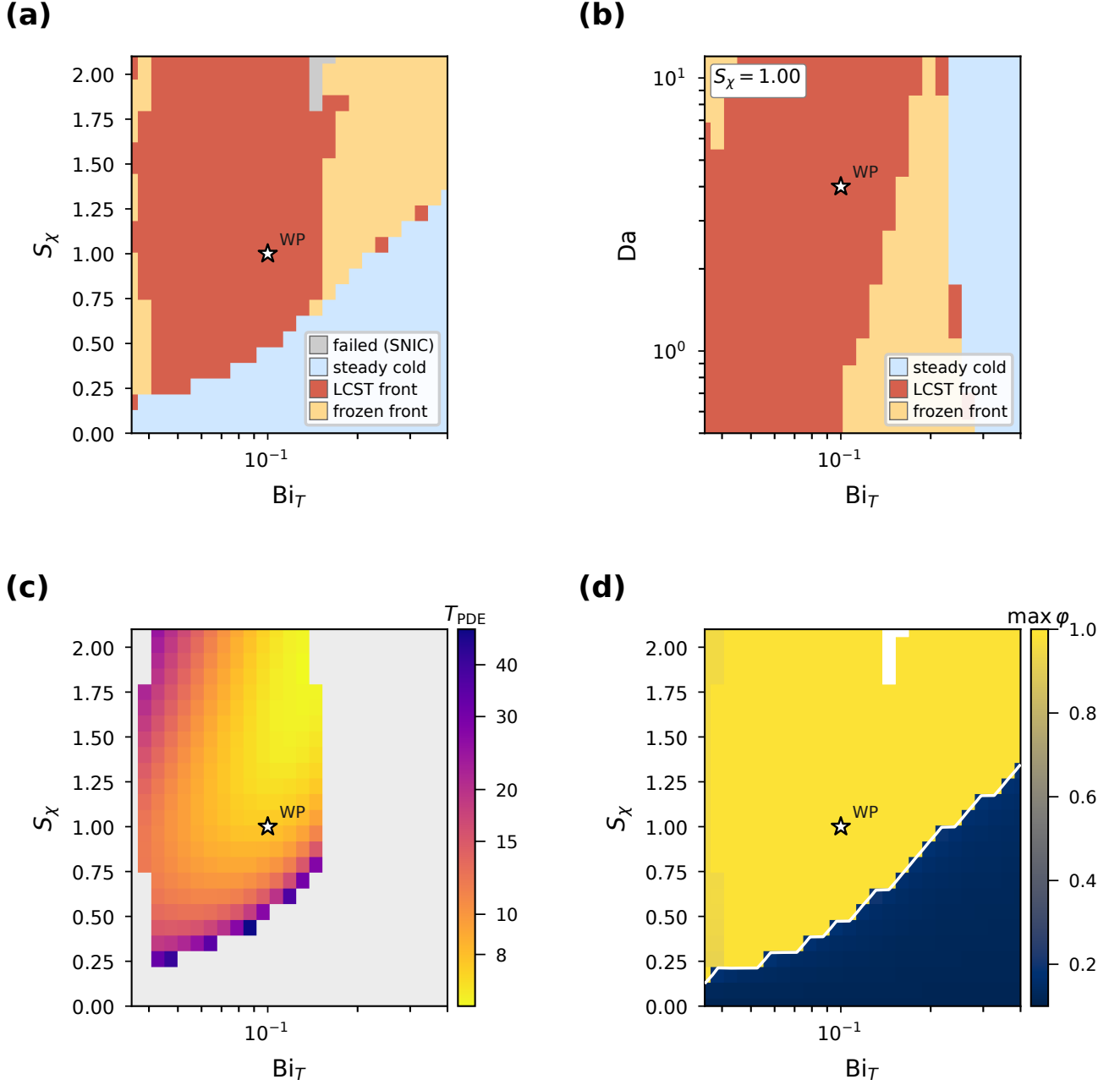


FIG. 2. Phase diagram of the realized PDE attractors at the default working-point reactivity, from a 25×25 default-IC sweep. Star: working point (WP); the bottom legend lists every regime present anywhere in panels (a)/(b). **(a)** Regime classification on (Bi_T, S_χ) . Light-grey cells (*failed*) mark integrator divergences along the saddle-node-on-invariant-circle (SNIC) boundary at large S_χ , where the cycle period diverges past the integration window. **(b)** Regime classification on (Bi_T, Da) at $S_\chi = 1.00$, confirming that the LCST-front regime persists over a decade in reactivity. **(c)** Surface oscillation period T_{PDE} in the LCST-front regime, drawn on a logarithmic colour scale to expose the SNIC divergence at the upper- and lower- S_χ boundaries. Light-grey background cells are non-oscillating (steady cold or frozen front); see (a) for their classification. **(d)** Peak polymer fraction $\max_{(\xi,t)} \varphi$. The white contour marks the LCST threshold $\varphi = 0.5$ and tracks the (a) cold-vs-front boundary, providing an independent check on the regime classifier.

the collapsed state ($J \sim 0.15$) with period $T \simeq 8\text{--}11$. The inner core ($\xi < 0.885$) stays at $J \simeq 1.39$ throughout the cycle. The temperature field tracks the asymmetry: θ_{surf} swings between ~ 1.5 and ~ 4.8 each cycle, while θ_{centre} is held nearly constant at $\theta_{\text{centre}} \simeq 3.2$ by heat conduction from the warm shell.

The frozen-front state [Fig. 3(g-i)] is stationary in time but strongly heterogeneous in space, with three regions: a super-swollen interior ($J \simeq 4$, $\varphi \simeq 0.04$) for $\xi \lesssim 0.91$, a narrow collapsed band at $\xi \simeq 0.92$ where J drops to $\simeq 0.15$ and φ rises to $\simeq 0.98$, and a super-swollen surface layer ($J \simeq 2.9\text{--}3.2$) for $\xi \gtrsim 0.93$. Two mechanisms combine to produce this profile. First, at $\text{Bi}_T = 0.18$ the surface heat flux is large enough to clamp the band's surface side below the re-ignition threshold, locking the cycle of Sec. IV B in phase 3. Second, at $S_\chi = 1.5$ the LCST coupling $S_\chi \theta \varphi^2$ pushes χ_{eff} so high that the only off-collapse solution of $\mu(J, \theta) = \mu_b$ is the dilute branch of the Flory–Huggins free energy, on which the elastic $\Omega_e(J - 1/J)$ contribution balances the reduced $\chi_{\text{eff}} \varphi^2$ at small φ . The textbook “two-zone” frozen front with a cold-swollen interior and a collapsed shell is therefore not realised in this model: the bulk either oscillates (LCST-front cycle) or migrates onto the same dilute branch that will reappear as the hot-runaway attractor in Sec. IV E. A fourth attractor, the hot-runaway hidden state, is reached only from non-default ICs and is analysed in Sec. IV E (see Fig. 8).

B. A relaxation oscillator on a spatially-generated slow manifold

The LCST-front cycle is a relaxation oscillator with two distinguishing features: (i) the fast variable is mechanical (J), the slow variables are thermal and chemical (θ , u); and (ii) the slow manifold is generated by the spatial structure of the gel rather than by an autonomous 0D nonlinearity. Each cycle traverses two stable branches of this manifold connected by fast jumps, and the period is dominated by the slowest of the four phases.

a. Fast-slow decomposition. At the working point the mechanical relaxation timescale set by the dimensionless mobility $\mathcal{M}_0$ is roughly two orders of magnitude shorter than the thermal ($1/\text{Bi}_T$) and reactant-supply ($1/\text{Bi}_c$) timescales. Numerically, the speed $|d(\theta_{\text{surf}}, J_{\text{surf}})/dt|$ along the limit cycle [Fig. 4(b)] varies by a factor of ~ 600 between the slow drifts and the fast jumps. We may therefore treat J as a fast variable and follow the standard relaxation-oscillator construction.

b. Spatially generated slow manifold. Quasi-static balance of the surface mechanical equation with the Robin boundary condition $\text{Bi}_\mu(\mu - \mu_b) = 0$ gives the locus

$$\mu(J, \theta) = \mu_b \quad (6)$$

in the (θ, J) plane. The Flory–Huggins–Rehner free energy is non-monotonic in J for sufficiently large $\chi_{\text{eff}} = \chi_\infty + S_\chi \theta + \chi_1 \varphi$ (above the LCST), so this locus has the

canonical S-shape [Fig. 4(a)]: an upper *swollen* branch ($J \gtrsim 0.8$) for $\theta \leq \theta_{\text{up}}$, a lower *collapsed* branch ($J \lesssim 0.2$) for $\theta \geq \theta_{\text{lo}}$, and a saddle (unstable middle branch) connecting the two folds. At the working point, $\theta_{\text{lo}} \simeq 0.85$ and $\theta_{\text{up}} \simeq 3.18$, defining a bistable region $\theta \in [\theta_{\text{lo}}, \theta_{\text{up}}]$ in surface temperature. Crucially, this slow manifold lives at the surface and is enforced by the surface BC; the bulk is buffered away from it by transport. The contrast is made visible in Fig. 4(b): the surface trajectory traverses the entire S-curve between the two folds, while the centre trajectory (inset) is a small loop confined to the swollen branch's interior at $(\theta \approx 3.24, J \approx 1.50)$, two orders of magnitude smaller in extent and never touching the manifold. A 0D homogeneous-state reduction has no analog of this manifold, since $\mu = \mu_b$ would have to be imposed everywhere simultaneously together with the heat and reactant balances (Sec. IV D).

c. Cycle traversal. The PDE limit cycle, projected onto $(\theta_{\text{surf}}, J_{\text{surf}})$, traces the slow manifold's two stable branches connected by near-vertical fast jumps [Fig. 4(b)]. We label four canonical phases:

1. **Ignition (slow, on swollen branch).** The surface starts at the cold end of the swollen branch ($\theta \simeq 1.5$, $J \simeq 1.4$) with abundant reactant ($u_{\text{surf}} \simeq 1$). A small thermal fluctuation amplifies the Arrhenius factor; the heat source $\text{Da} J \hat{R}$ exceeds the surface heat loss $\text{Bi}_T \theta_{\text{surf}}$ and θ_{surf} rises monotonically.
2. **Collapse (fast).** As θ_{surf} exceeds the upper-fold value, the swollen branch ceases to exist; J falls rapidly along the fast manifold from $J \sim 0.8$ to $J \sim 0.15$ over a small fraction of the cycle period.
3. **Quenching (slow, on collapsed branch).** With $\varphi \sim 1$ in the collapsed shell the diffusion factor $D(\varphi) \propto (1 - \varphi)^2$ drops by three orders of magnitude and the catalyst-accessibility factor $(1 - \varphi)^{m_{\text{act}}}$ by ten or more, throttling reactant supply and quenching the chemistry. The Robin BC pins $u(\xi = 1)$ near the bath value, but inside the closed shell the reactant field is several decades lower [Fig. 4(c)] and the heat source effectively vanishes; θ_{surf} then relaxes towards the bath temperature on the surface heat-loss timescale $1/\text{Bi}_T$. Phase 3 is generally the longest phase of the cycle.
4. **Re-swelling (fast).** When θ_{surf} falls below the lower-fold value, the collapsed branch terminates and J jumps back up to the swollen branch. The diffusion barrier reopens, the bath rapidly resupplies reactant, and the cycle returns to phase 1.

The slow drifts (phases 1 and 3) and the fast jumps (phases 2 and 4) are resolved on the limit cycle as differently-coloured segments [Fig. 4(b)] and on the reactant kymograph [Fig. 4(c)] as alternating bands of high and low u at the surface. The corresponding $J(\xi, t)$ and $\theta(\xi, t)$ fields at the working point are shown explicitly in Fig. 3(d,e).

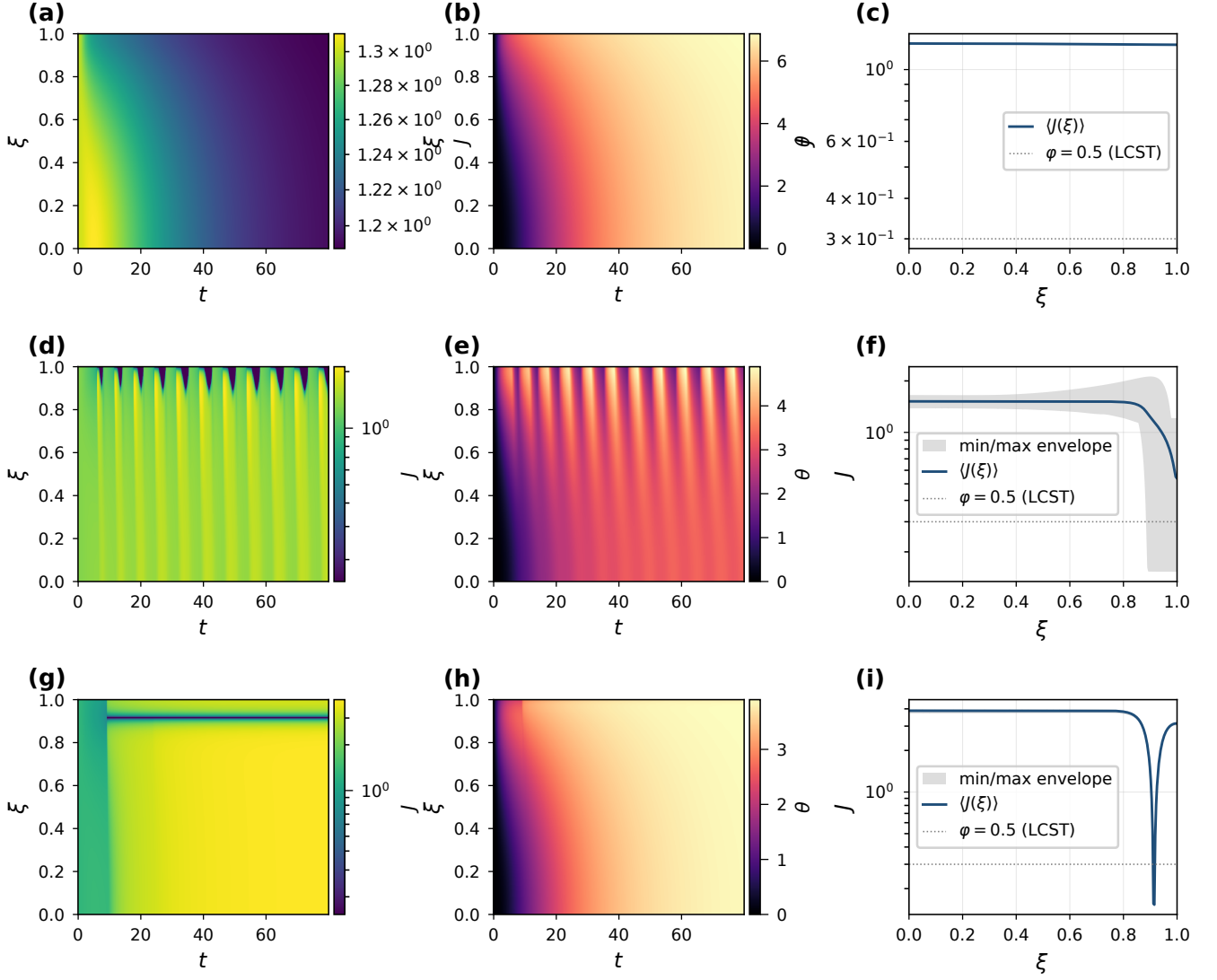


FIG. 3. Spatiotemporal structure of the three default-IC attractors, organised into three rows of three panels each. Within every row the columns show, from left to right, the $J(\xi, t)$ kymograph, the $\theta(\xi, t)$ kymograph, and the time-mean spatial profile $\langle J(\xi) \rangle_t$ with a grey min/max envelope band; the grey dotted horizontal line marks the LCST threshold $J = \varphi_{p0}/0.5 \simeq 0.30$, so any region of the profile dropping below it has crossed LCST. All three rows show $t \in [0, 80]$ so the relaxation from the default cold IC onto each attractor is visible in the kymographs; the time-mean profile in the right column is computed over the post-transient sub-window $t \in [40, 80]$ to avoid IC bias. (a–c) Cold-swollen steady state at $\text{Bi}_T = 0.10$, $S_\chi = 0.20$; (b) shows θ rising from the IC value $\theta_0 = 0$ to the homogeneous reaction–cooling balance $\theta \simeq 6.9$ within $t \lesssim 30$, while φ stays well below the LCST everywhere. (d–f) LCST-front oscillation at the working point ($\text{Bi}_T = 0.10$, $S_\chi = 1.00$); cycles develop from the cold IC and reach stable amplitude by $t \sim 30$. The envelope in (f) widens from a thin core ($\xi < 0.885$, where the trace stays above LCST) to the surface shell, where J swings between ~ 1.4 and ~ 0.15 each cycle. (g–i) Frozen partial-collapse front at $\text{Bi}_T = 0.18$, $S_\chi = 1.50$; the collapsed band forms within $t \sim 20$ and locks at $\xi \simeq 0.92$. Panel (i) shows the steady three-region profile (super-swollen interior, thin collapsed band, super-swollen surface layer); see Sec. IV A for the mechanism. The fourth attractor (hot-runaway hidden state) is accessed only from non-default ICs and is shown in Fig. 8.

d. Self-limiting collapse. The same φ -dependent diffusion barrier that quenches the reaction at the surface during phase 3 also prevents collapse from invading the gel interior. The LCST front penetrates only to $\xi_{\text{LCST}} \simeq 0.885$ at the working point; the inner 88.5% of the slab never crosses the LCST and remains swollen throughout the cycle. This barrier explains why the

classifier- allowed regimes *globally collapsed steady* and *globally collapsing oscillation* are absent from the 1000 default-IC simulations of Sec. IV A: once any portion of the gel collapses, the inner region is shielded from further reactant supply and cannot ignite. The same mechanism is, in this sense, a self-limiting protective feature of this class of thermo-LCST gel against runaway global

collapse despite the strongly exothermic Arrhenius kinetics [16, 18].

e. Period structure. Across the LCST-front regime [Fig. 2(c), Fig. 4(c)], the cycle period T_{PDE} takes its minimum value $\simeq 6$ in the upper portion of the LCST-front region just inside the SNIC exit, and diverges at both the upper- and lower- S_χ boundaries, where the cycle is destroyed through a saddle-node-on-invariant-circle (SNIC) bifurcation analyzed in Sec. IV D. The cooling timescale $1/\text{Bi}_T$ sets the order of magnitude of the period, but T_{PDE} is not a single power law of Bi_T : phase 3 is also controlled by the residual interior heat conduction and by the timescale at which the surface BC restores u_{surf} after the diffusion barrier reopens.

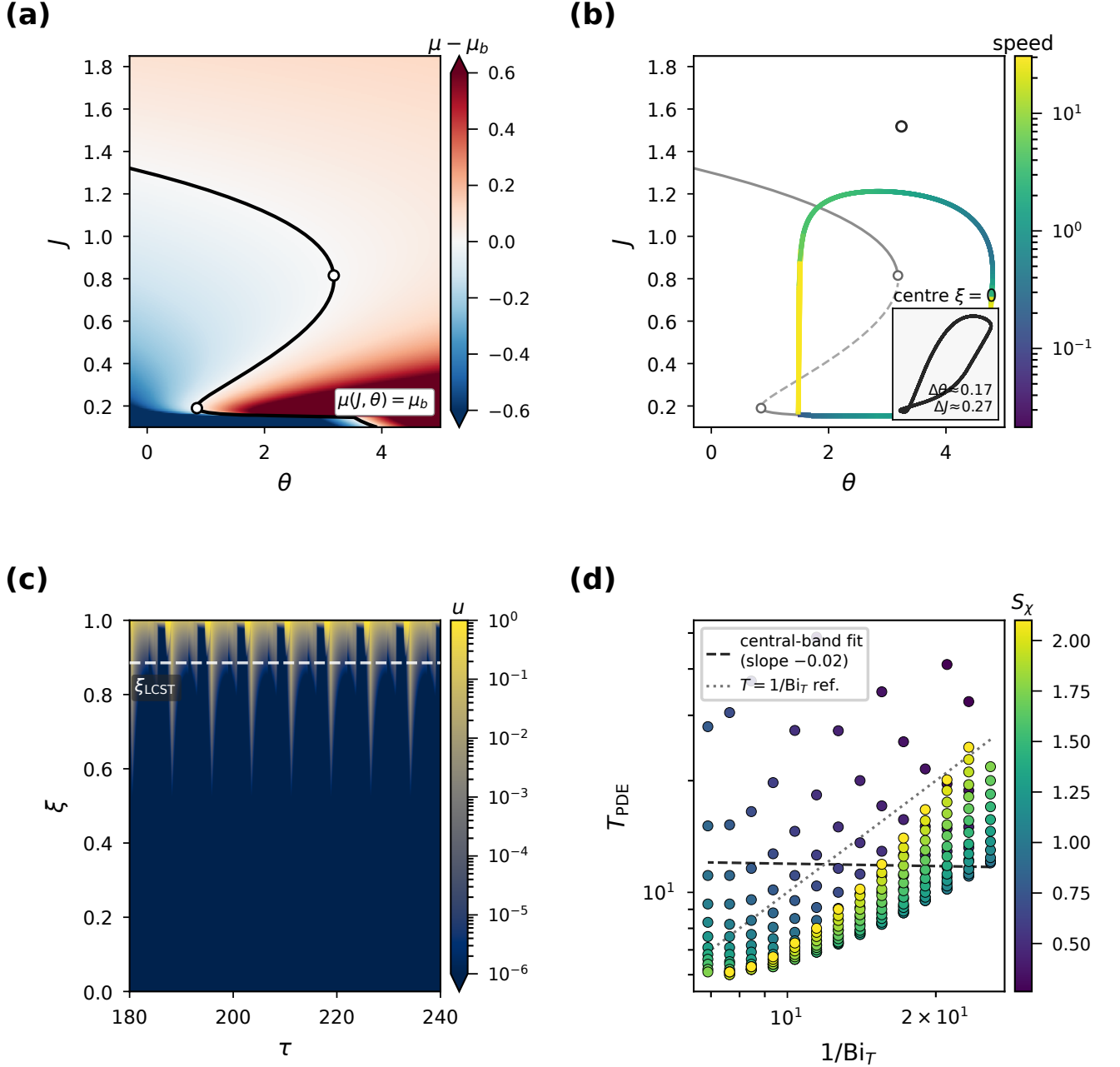


FIG. 4. Anatomy of the LCST-front relaxation cycle at the working point. **(a)** Slow-manifold geometry. Heatmap of $\mu(J, \theta) - \mu_b$ in the (θ, J) plane (saturated at ± 0.6 to expose the relevant region) with the black contour $\mu(J, \theta) = \mu_b$ overlaid; this contour is the slow manifold, a level set of the Flory–Huggins–Rehner chemical potential at the bath value, and is enforced at $\xi = 1$ by the Robin BC. Open circles mark the upper and lower folds. **(b)** PDE limit cycles in the same coordinates. The surface trajectory ($\xi = 1$, speed-coloured on a log scale) traverses the full S-curve between the folds, with $\sim 600\times$ variation between the slow drifts on the swollen/collapsed branches and the near-vertical fast jumps; the four phases (ignition, collapse, quenching, re-swelling) are read off directly from the cycle topology and segment colour. The grey faint S-curve is reproduced from (a) as a reference. The inset shows the centre trajectory ($\xi = 0$): a small loop confined to the swollen branch’s interior, two orders of magnitude smaller than the surface cycle. The contrast is the §IV.B “spatially generated slow manifold” argument made visible: the manifold is a surface phenomenon, the bulk is buffered away from it. **(c)** Reactant kymograph $u(\xi, \tau)$ on a log scale over five cycles. The thin bright band at $\xi \rightarrow 1$ is the surface boundary layer pinned by the Robin BC near the bath value; inside the collapsed shell the field is several decades lower because reactant cannot replenish past the closed diffusion barrier. The white dashed line at $\xi_{\text{LCST}} \approx 0.885$ is the innermost ξ that ever crosses the LCST; below it lies the polymer- shielded core that the front never invades. **(d)** Period scaling T_{PDE} vs $1/\text{Bi}_T$ across all oscillating cells; colour encodes S_χ . Dashed line: log-log fit on the central band $S_\chi \in [0.30, 1.05]$; dotted: the reference $T = 1/\text{Bi}_T$. The cooling timescale sets the order of magnitude but is not a single power-law collapse variable because phase 3 also depends on the diffusion-barrier release time.

C. Scaling laws for the LCST-front cycle

The relaxation-oscillator picture of Sec. IV B predicts four observable scaling laws. We verify three of them quantitatively and reframe a fourth qualitatively against PDE simulations at $N=301$ over the working point and its neighbourhood. Throughout we use the fold values $\theta_{lo} \approx 0.844$ and $\theta_{up} \approx 3.184$, obtained from the Flory–Huggins–Rehner chemical potential at the bath value $\mu_b = \mu(J_{init}, 0) \approx 0.066$, together with the dimensionless penetration depths $L_T \equiv \sqrt{\alpha/\text{Bi}_T}$ and $L_c \equiv \sqrt{\delta/\text{Bi}_c}$.

a. Period. Tikhonov’s theorem for singularly perturbed systems [21, 22] predicts that, in the limit $\text{Bi}_T \rightarrow 0$, the cycle period is dominated by the slow thermal relaxation between the two folds of the slow manifold and the product $T_{\text{PDE}} \cdot \text{Bi}_T$ approaches the universal constant

$$T_{\text{PDE}} \cdot \text{Bi}_T \rightarrow \ln(\theta_{up}/\theta_{lo}) \approx 1.328. \quad (7)$$

Figure 5(a) shows $T_{\text{PDE}} \cdot \text{Bi}_T$ across the $(\text{Bi}_T, S_\chi) \in [0.04, 0.12] \times [0.9, 1.5]$ grid. The oscillating cells collapse to (7) with maximum deviation 13.5%; the residual rises monotonically with Bi_T , consistent with sub-leading $\mathcal{O}(\text{Bi}_T)$ corrections from the fast jump phases that the singular limit suppresses asymptotically [23].

b. Amplitude. The same fast–slow decomposition predicts an amplitude set by the fold spacing alone,

$$h_{\text{eff}} \equiv \Delta\theta_{\text{surf}} = \theta_{up} - \theta_{lo} \approx 2.340, \quad (8)$$

independent of Bi_T and S_χ to leading order. Figure 5(b) shows the measured surface temperature amplitude at six oscillating points across $(\text{Bi}_T, S_\chi) \in \{0.05, 0.10, 0.15\} \times \{0.7, 0.8, 1.0, 1.1, 1.3\}$. All six lie within 13.4% of (8); the band is roughly symmetric about the prediction, with no systematic drift versus either control parameter.

c. Front depth. The front depth—the dimensionless distance from the surface at which the LCST crossover $\varphi = 0.5$ never penetrates—reports the cycle’s spatial reach.

Physical image. During the hot phase a deswelling front nucleates at the surface and advances inward at the speed set by whichever transport process is more sluggish: thermal diffusion (depth L_T), which determines how far the surface temperature spike penetrates before being radiated back to the bath, or reactant supply (depth L_c), which limits the depth over which exothermic combustion can sustain itself. Whichever penetration length is shorter pins the innermost stable front position.

Derivation. Balancing the local reaction–diffusion fluxes at the deepest point reached gives, at leading order in φ^m with $m = \min(m_{\text{act}}, m_{\text{diff}})$,

$$1 - \xi_{\text{LCST}} = c \cdot \min(L_T, L_c), \quad (9)$$

with c a geometric prefactor of order unity. The derivation implicitly assumes that c is regime-independent—the same constant should appear whether L_T or L_c is the limiting scale. We test this assumption numerically.

L_c -branch verification. Phase D scans a $(m_{\text{act}}, m_{\text{diff}}) \in \{1, \dots, 7\}^2$ grid at the working point $(\text{Bi}_T, \text{Da}, S_\chi) = (0.10, 4, 1.0)$ with default $(\alpha, \delta, \text{Bi}_c) = (0.20, 0.08, 0.70)$, giving $L_c = 0.338$ as the limiting scale throughout. Figure 6(a) plots $(1 - \xi_{\text{LCST}})/L_c$ versus $M \equiv m_{\text{act}} + m_{\text{diff}}$. Once $M \geq 6$, where the LCST barrier $(1 - \varphi)^m$ becomes sharp enough to support a well-defined front, the ratio plateaus at $c_c = 0.388 \pm 0.002$ ($\sigma/c = 2.5\%$ across 15 cells), in clean agreement with (9). Figure 6(b) is the same data colour-coded as a five-way attractor map (cycle / overswollen-front / frozen-front / hot-runaway / cold-SS); the saturated cohort sits squarely inside the cycle region.

L_T -branch and the prefactor question. Entering the L_T -limited regime requires $\alpha/\text{Bi}_T < \delta/\text{Bi}_c$ while preserving the LCST-front cycle. We find this constraint nearly incompatible with cycle survival: at the patch parameters $(\alpha, \delta, \text{Bi}_c) = (0.04, 0.50, 0.70)$, only the point $\text{Bi}_T = 0.10$ sustains the cycle, giving $c_T \approx 0.20$ —distinct from the L_c -branch value $c_c \approx 0.39$. This suggests that the two limits are governed by different geometric constants: reactant exhaustion and thermal screening pin the front via distinct local balances, and a unified single-prefactor form $1 - \xi_{\text{LCST}} = c \min(L_T, L_c)$ cannot be claimed from current data. We leave a systematic exploration of the L_T -branch prefactor to a follow-up.

Corollary. As a by-product, sustained LCST-front cycles appear to require $L_T \gtrsim \mathcal{O}(1)$: every attempt to shrink α or raise δ enough to push L_T well below L_c destroys the cycle by collapsing it to a frozen-front or uniformly-deswollen steady state. This is an existence constraint on the LF cycle not visible in the (Bi_T, S_χ) phase diagram of Sec. IV A.

d. Onset. A standard 0D analysis of the homogeneous swollen branch gives a loss-of-stability threshold

$$\text{Da}_c^{0D} = \text{Bi}_T / (J_{\text{init}} \Gamma_A), \quad (10)$$

linear in Bi_T with prefactor of order unity. The PDE onset Da_c^{PDE} , measured by 8-iteration bisection at four $\text{Bi}_T \in \{0.06, 0.10, 0.16, 0.25\}$, lies above (10) by a factor of order ten and tracks Bi_T with $R^2 = 0.88$ but with a prefactor mismatch. Diagnostics at the bisection bracket reveal an intermediate “over-swollen-front” attractor sitting between the cold-swollen SS and the LF cycle, with $\langle J \rangle \approx 3.8$ and a thin self-trapped LCST layer at the surface. The true onset mechanism is therefore a saddle-node of the cold-swollen *spatial* branch rather than a Hopf bifurcation of the 0D reduction [16, 24]. The qualitative prediction—onset is controlled by Bi_T rather than Da alone—survives, but (10) should not be used for quantitative onset prediction.

In summary, three of the four predicted scaling laws—period, amplitude, and front-depth saturation in the L_c -limited branch—verify cleanly within 13% of their leading-order forms, while the onset law is qualitatively correct but quantitatively reframed by the PDE saddle-node. The L_T -limited branch of the front-depth scaling is left open as a follow-up.

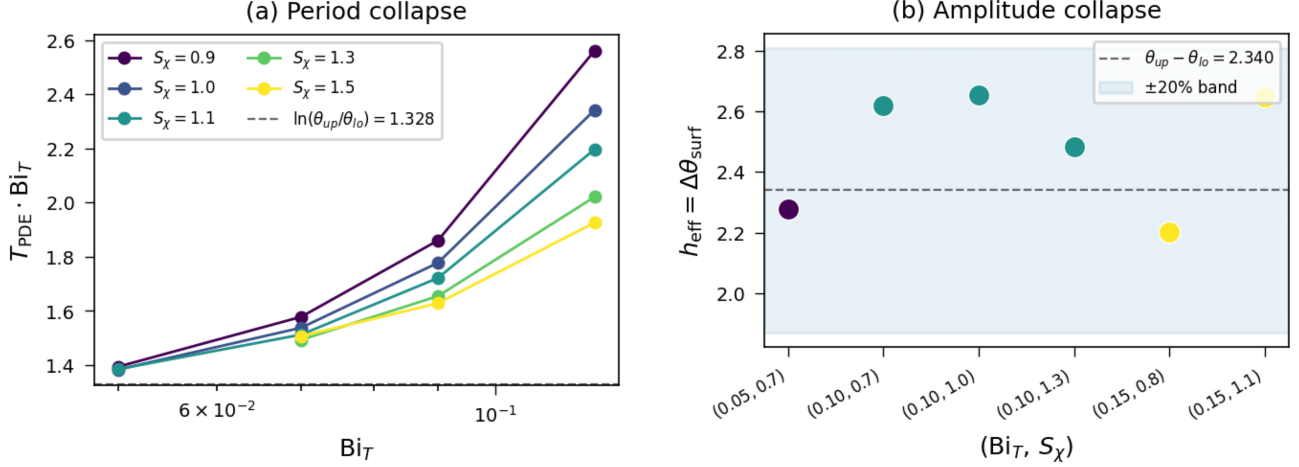


FIG. 5. Period and amplitude scalings of the LCST-front cycle. **(a)** $T_{\text{PDE}} \cdot \text{Bi}_T$ vs Bi_T at five $S_\chi \in [0.9, 1.5]$. Dashed line: $\ln(\theta_{\text{up}}/\theta_{\text{lo}}) = 1.328$ from the F–H–R fold values. Eighteen oscillating cells collapse to within 13.5% of the singular-limit constant. **(b)** Measured amplitude $h_{\text{eff}} = \Delta\theta_{\text{surf}}$ at six (Bi_T, S_χ) points. Dashed line: prediction $\theta_{\text{up}} - \theta_{\text{lo}} = 2.340$; shaded band: $\pm 20\%$.

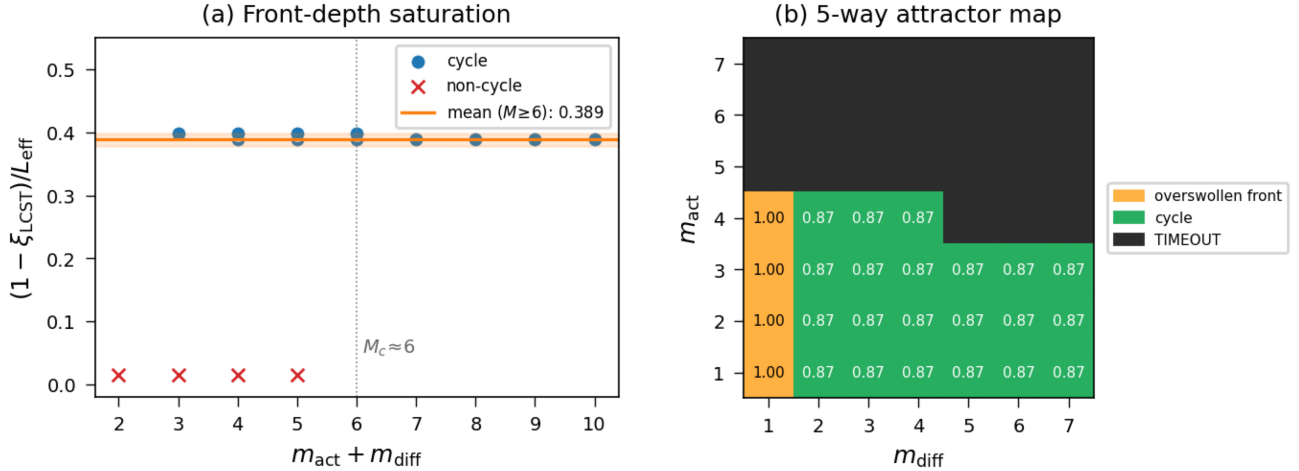


FIG. 6. Front-depth saturation and the LF cycle's parameter envelope. **(a)** $(1 - \xi_{\text{LCST}})/L_{\text{eff}}$ versus $M = m_{\text{act}} + m_{\text{diff}}$ across the $(m_{\text{act}}, m_{\text{diff}})$ main grid at the working point. Blue circles: cycle cells; red crosses: non-cycle (in the overswollen-front column $m_{\text{diff}} = 1$). Once $M \geq M_c \approx 6$, the ratio plateaus at 0.388 ± 0.002 ($\sigma/c = 2.5\%$), confirming (9) in the L_c -limited branch ($L_{\text{eff}} = L_c = 0.338$ at these parameters). A single point in the L_T -limited regime ($\alpha = 0.04$, $\text{Bi}_T = 0.10$; not shown) gives the distinct ratio ≈ 0.20 , suggesting that the two branches have different geometric prefactors. **(b)** Five-way attractor classification on the same grid. Cells are coloured by their long-time attractor type; numbers show ξ_{LCST} in cycle and frozen-front cells. Dark cells timed out; the saturation argument is established on the $M \in [6, 10]$ cohort that did converge.

D. Linear stability is necessary but not sufficient

Section IV B showed that the LCST-front attractor is a relaxation cycle whose slow manifold is generated by the surface boundary condition together with the φ -dependent diffusion barrier. No spatially uniform reduction can produce that manifold: at the surface the lo-

cus $\mu(J, \theta) = \mu_b$ is enforced by the Robin BC, while in the interior the same locus is irrelevant because the bulk equilibrium is buffered by transport. Linear stability of an available homogeneous branch is therefore at best a *necessary* condition for the PDE to oscillate; it cannot be sufficient because the cycle traverses two branches of a slow manifold that does not exist in the 0D reduction.

This subsection makes the necessary-condition statement quantitative on two complementary axes: (i) where the homogeneous SS even exists as a real root, and (ii) how the cycle is destroyed at the parameter- plane boundaries of the LCST-front regime.

a. Existence of the homogeneous swollen branch (analytic). The 0D steady-state system (4) admits an explicit parameterization: at any pair (J, θ) with $J > \varphi_{p0}$, the two SS conditions solve in closed form for the dependent control parameters,

$$S_\chi(J, \theta) = \frac{\mu_b - \mu_{\text{base}}(J, \theta)}{\theta \varphi^2}, \quad \text{Bi}_T(J, \theta) = \frac{K(J, \theta)}{\theta (1 + K/\text{Bi}_c)}, \quad (11)$$

where μ_{base} is the part of μ that excludes the $S_\chi \theta \varphi^2$ coupling and $K(J, \theta) = \text{Da } J (1 - \varphi)^{m_{\text{act}}} T(\theta)$. The entire 0D SS surface is therefore the image of the (J, θ) plane under (11), and the saddle-node fold locus in (Bi_T, S_χ) is exactly the image of the curve $\det \partial(F_1, F_2)/\partial(J, \theta) = 0$. Numerically that locus is a single connected component in (J, θ) ; under the map (11) it self-folds in (Bi_T, S_χ) into two branches meeting at a Whitney cusp at $(\text{Bi}_T, S_\chi) \simeq (0.43, 2.17)$ — a cusp catastrophe in the homogeneous-SS surface.

Counting the real roots of (4) cell-by-cell on a 360×240 grid in (Bi_T, S_χ) via the closed-form inverse (11) (one sign-change scan per cell along J , exact and $\sim 10^2 \times$ faster than fsolve) gives Fig. 7(a). The displayed plane decomposes into $\sim 37\%$ *swollen-only*, $\sim 51\%$ *collapsed-only*, and $\sim 11\%$ *bistable* (3 roots: 2 stable plus 1 saddle); exactly 0% of cells have 2 roots. This odd-only multiplicity is forced by the fold topology — any horizontal line in (Bi_T, S_χ) at fixed S_χ crosses the two fold arcs an even number of times, giving root counts 1, 3, 5, $\dots$. The default working point $(\text{Bi}_T = 0.10, S_\chi = 1.0)$ and the canonical C-cell $(\text{Bi}_T = 0.059, S_\chi = 1.80)$, both used elsewhere in this section, sit deep inside the “collapsed-only” region: at neither point does the swollen homogeneous SS that one would linearise about exist as a real root, so a 0D LSA about the swollen branch is structurally unavailable. The PDE nevertheless finds the LCST-front cycle there. The mismatch is the necessary-but-not-sufficient statement made geometric: the spatially-generated slow manifold of Sec. IV B continues to exist past the saddle-node fold of the homogeneous problem because its swollen branch is enforced *at the surface* by the Robin BC, not required to coexist with bulk reactant balance.

b. SNIC scaling at the upper- S_χ exit. At the upper- S_χ boundary of the LCST-front regime [Fig. 2(c), Fig. 4(d)], the cycle period diverges as $S_\chi \rightarrow S_\chi^{(c)}$ from below. We test the period scaling along a 1D slice through this boundary against the two canonical infinite-period scenarios

$$T_{\text{SNIC}}(S_\chi) \propto (S_\chi^{(c)} - S_\chi)^{-1/2}, \quad T_{\text{hom}}(S_\chi) \propto -\ln(S_\chi^{(c)} - S_\chi), \quad (12)$$

using the same 1D scan [Fig. 7(b)]. A two-parameter least-squares fit of $T_{\text{PDE}}(S_\chi)$ on the bracket

$S_\chi \in [1.10, 1.32]$ yields a SNIC residual sum of squares $\text{RSS}_{\text{SNIC}} = 0.068$ versus $\text{RSS}_{\text{hom}} = 0.152$, with the SNIC fit giving the critical value $S_\chi^{(c)} \approx 1.34$. The SNIC inverse-square-root law captures the data more than twice as well as the homoclinic log law, consistent with the cycle being destroyed by the collision of a saddle and a node on the limit cycle rather than by a homoclinic orbit hitting a saddle off the cycle. In either reading the cycle’s disappearance is a global topological event in the surface $(\theta_{\text{surf}}, J_{\text{surf}})$ slow-manifold plane and not a local linear bifurcation of any single homogeneous branch — the expected signature of a relaxation oscillator on a spatially generated slow manifold.

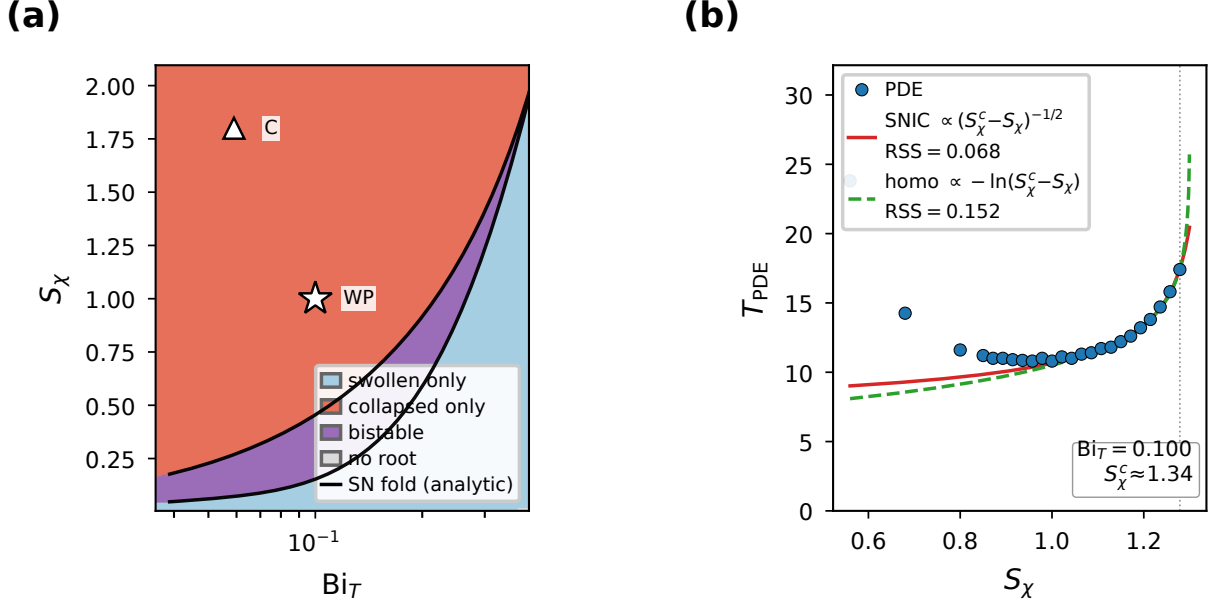


FIG. 7. Necessary-but-not-sufficient evidence on (Bi_T, S_χ) . **(a)** Real-root multiplicity of the homogeneous-SS system (4), computed by the closed-form parameter inversion (11) (1D sign-change scan along J , 360×240 pixels). Colours: swollen only (sky blue), collapsed only (terra cotta), bistable / 3 roots (purple), no root in the displayed window (light grey). The black analytic curve is the saddle-node fold locus, the image of $\det \partial(F_1, F_2)/\partial(J, \theta) = 0$ under (11); it self-folds into two arcs meeting at a Whitney cusp at $(\text{Bi}_T, S_\chi) \simeq (0.43, 2.17)$. White star: working point; white triangle: C-cell. **(b)** Period-divergence scaling at the upper- S_χ exit. Open circles: PDE cycle period along a 1D scan in S_χ at fixed Bi_T . Solid line: SNIC fit $T \propto (S_\chi^{(c)} - S_\chi)^{-1/2}$ ($S_\chi^{(c)} \approx 1.34$, RSS = 0.068). Dashed line: homoclinic fit $T \propto -\ln(S_\chi^{(c)} - S_\chi)$ (RSS = 0.152, same window). The SNIC law captures the divergence $\sim 2\times$ better than the homoclinic law.

c. Summary. The 0D linear stability analysis carries a precise but limited informational content: it identifies the local stability of available homogeneous branches, but the swollen homogeneous branch fails to exist over $\sim 51\%$ of the parameter plane covered here, and even where it does the cycle that the PDE realises is generically not the small-amplitude continuation of any homogeneous Hopf bifurcation. The dominant nonlinear behaviour of this thermo-LCST gel is a relaxation oscillator on a spatially generated slow manifold, born and destroyed by global topological events (Sec. IV B and Eq. (12)). Whether the same parameter plane harbours a hidden second attractor not visible to either linear analysis or default-IC numerical scans is settled in Sec. IV E.

E. A hidden hot-runaway state and the IC-induced hysteresis prediction

Section IVD closed by asking whether the collapsed-only region of the parameter plane—where the swollen homogeneous SS does not exist yet the PDE oscillates—corresponds to genuine attractor multiplicity of the spatially extended problem. We resolve this by initial-condition (IC) sweeps at a representative point

in this region, the canonical C-cell at $\text{Bi}_T = 0.059$, $S_\chi = 1.80$.

a. Two co-existing attractors. At identical bath conditions and reaction parameters, simulations initialised in different regions of (J_0, θ_0) space converge to one of two qualitatively distinct attractors [Fig. 8(c)]: (i) the LCST-front cycle of Sec. IV B, with bulk-mean $\langle J \rangle \approx 2$, $\langle \theta \rangle \approx 2$, and (ii) a previously-unreported *hot-runaway* steady state with $\langle J \rangle \approx 5$, $\langle \theta \rangle \approx 12$. The two attractors differ by $\Delta \langle J \rangle \approx 3.3$ and $\Delta \langle \theta \rangle \approx 10$, far beyond any numerical tolerance, so the bistability is genuine.

b. Spatial structure and mechanism of the hot-runaway state. The hot-runaway state [Fig. 8(b)] is spatially uniform with $J \simeq 5.2$, $\theta \simeq 12$, $\varphi \simeq 0.026$, and a thin reaction layer at the surface where reactant is consumed almost entirely ($u_{\text{surf}} \sim 10^{-7}$). The mechanism is diffusion-limited combustion: the high- θ Arrhenius factor exceeds 10^5 and burns through any reactant in the boundary layer, but the surface BC continues to deliver reactant at the diffusion-limited rate Bi_c . The gel remains super-swollen because the high- J Flory-Huggins minimum coexists with the collapsed minimum whenever χ_{eff} is sufficiently large; at $\varphi \sim 0.026$ the $\chi_{\text{eff}} \varphi^2$ penalty is negligible and the elastic term $\Omega_e (J - 1/J)$ becomes the dominant contribution to the chemical po-

tential. The choice between the two Flory–Huggins minima is set by the IC: trajectories landing in the basin of the swollen branch reach hot-runaway, while trajectories landing in the basin of the collapsed branch are ejected to the LCST-front cycle by the surface BC and the diffusion-barrier feedback.

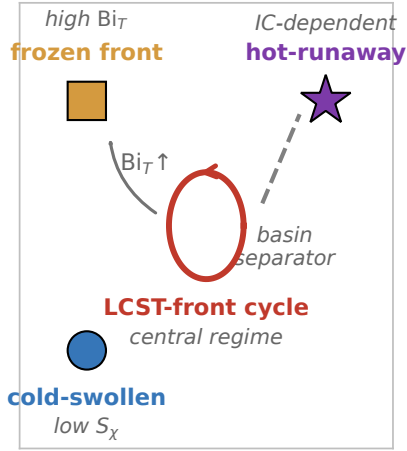
c. Basin asymmetry and why the default sweep misses the runaway. A refined IC sweep on a 11×11 grid at the C-cell [Fig. 8(c)] places the basin boundary at roughly $J_0 \approx 1.0$, $\theta_0 \approx 1$: only ICs in the upper-left quadrant of (J_0, θ_0) space (cold, swollen) reach the cycle. ICs with any of (i) $J_0 \lesssim 0.4$ (pre- collapsed), (ii) $\theta_0 \gtrsim 3$ (ignited), or (iii) combinations spanning the basin boundary fall into the runaway. The basin of the cycle covers $\sim 13\%$ of the IC grid (16 of the 119 successfully integrated cells of an 11×11 grid spanning $J_0 \in [0.20, 1.30]$, $\theta_0 \in [0, 10]$), a small fraction of the total. This asymmetry explains why the default cold-IC sweep of Fig. 2 sees only the cycle and never the runaway: the default cold IC sits well inside the cycle’s basin at every grid point of the parameter sweep.

d. IC-induced hysteresis: a falsifiable prediction. A 1D slice through the basin space gives a quantitative, experimentally accessible signature of the bistability [Fig. 8(d)]. Holding all reaction and bath parameters fixed at the C-cell value with the initial reactant loading $u_0 = 0.5$, we vary the initial thermal pulse θ_0 from 0 to 6 for two seed conditions:

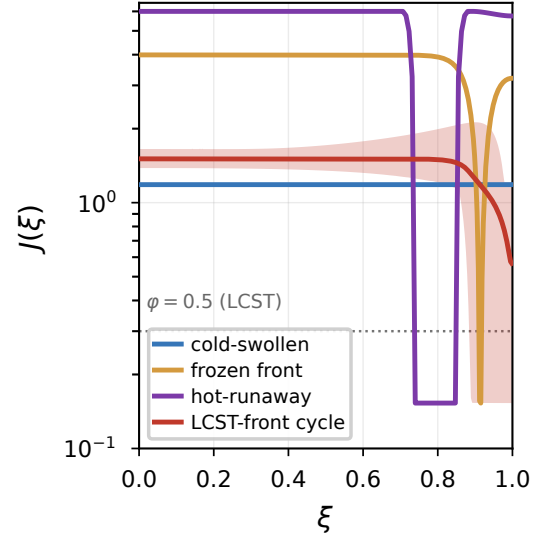
- “cold” seed ($J_0 = 1.30$): initially near the cold-swollen homogeneous state. For θ_0 below the threshold $\theta_0^* \approx 2.4$ the trajectory remains in the cycle’s basin and $\langle J \rangle_{\text{late}} \approx 2.0$; for θ_0 above the threshold the seed crosses the basin boundary and $\langle J \rangle_{\text{late}} \approx 5.9$, indistinguishable from the runaway state. The transition is sharp—the discontinuity in $\langle J \rangle_{\text{late}}$ falls within a single $\Delta\theta_0 = 0.25$ sampling step.
- “pre-collapsed” seed ($J_0 = 0.20$): initialised inside the runaway basin; for every θ_0 in the scanned range the trajectory terminates in the runaway-type state with $\langle J \rangle_{\text{late}} \approx 4.2\text{--}5.5$, independent of θ_0 .

The split between the two terminal-state curves over the bistable θ_0 interval $[0, 2.4]$ is the hysteresis signature. Experimentally, this would manifest as a sharp jump in long-time gel volume (or temperature) when the initial perturbation magnitude is varied across θ_0^* , with no return jump on the reverse sweep so long as the C-cell parameters are held fixed. This is a quantitative prediction of the model that the linear theory does not provide and that distinguishes spatially-extended thermo-LCST gels from the well-mixed CSTR analog [18].

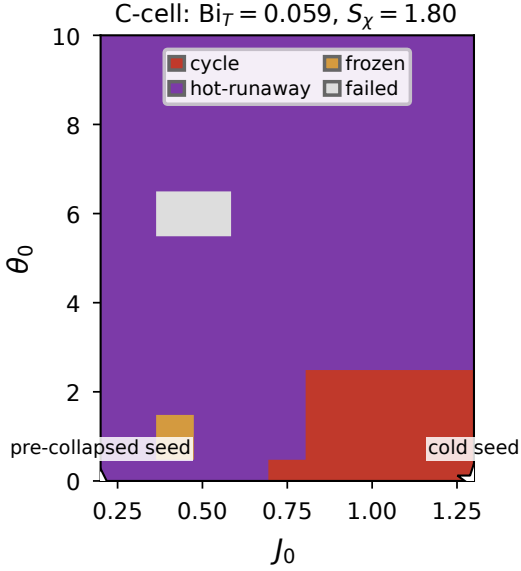
(a)



(b)



(c)



(d)

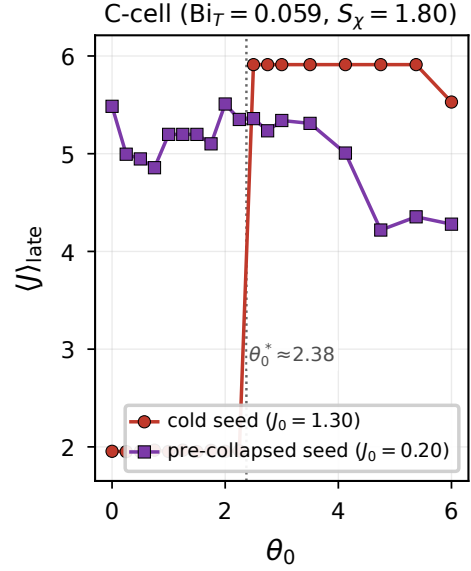


FIG. 8. Manifold portrait of the four PDE attractors and IC- space evidence at the C-cell. (a) Schematic state-space topology: the LCST-front cycle (centre) connects under increasing Bi_T to the frozen front; the cold-swollen and hot-runaway fixed points sit in parameter regions disjoint from the cycle, with the hot-runaway accessible only across the basin separator at the C-cell. (b) Late-time spatial profile $J(\xi)$ for one representative trajectory per attractor at its parameter point. The LCST-front cycle is shown as the envelope between $J_{\min}(\xi)$ and $J_{\max}(\xi)$ (red shading) plus the time-mean profile (red curve). The dotted line marks the LCST threshold $\varphi = 0.5$, i.e. $J = \varphi_{p0}/0.5 = 0.30$. (c) Refined basin of attraction at the C-cell ($\text{Bi}_T = 0.059$, $S_\chi = 1.80$). Each pixel of the 11×11 grid in (J_0, θ_0) space (fixed $u_0 = 0.5$) is coloured by the long-time attractor reached: red = LCST- front cycle, purple = hot-runaway steady state, gold = frozen partial collapse, grey = solver failure. The basin of the cycle covers $\sim 13\%$ of the IC grid; the runaway dominates, so the default cold-IC sweep of Fig. 2 captures only the cycle. (d) IC-induced hysteresis. Long-time bulk-mean $\langle J \rangle_{\text{late}}$ vs the initial thermal pulse θ_0 for two seed values of J_0 . Cold seed ($J_0 = 1.30$, red) jumps sharply from the cycle to the runaway at the basin threshold $\theta_0^* \approx 2.4$, within a single $\Delta\theta_0 = 0.25$ step; pre-collapsed seed ($J_0 = 0.20$, purple) terminates in the runaway state for every θ_0 . The split between the two terminal-state curves over the bistable interval is the experimentally accessible hysteresis signature.

The PDE attractor diagram is therefore richer than the single-IC sweep of Fig. 2 suggests, and richer than 0D LSA can predict. The hot-runaway state is invisible to both linear analysis and to default-IC numerical scans, and its appearance is contingent on the structure of the basin of attraction in initial-condition space. Combined with the slow-manifold relaxation cycle of Sec. IV B, this completes a four-attractor inventory of the spatially extended problem (cold-swollen, LCST-front cycle, frozen front, hot-runaway), whose existence and stability are governed not by any single homogeneous-state linearisation but by the spatial coupling between the Flory–Huggins free energy and the φ -dependent diffusion barrier.

V. CONCLUSION

The dominant attractor of the thermo-LCST gel slab is a relaxation oscillator whose slow manifold is generated by a spatial structure absent from any 0D reduction. The surface boundary condition pins the fast variable J to the locus $\mu(J, \theta) = \mu_b$, while the φ -dependent diffusion barrier $D_0(1 - \varphi)^{m_{\text{diff}}}$ sets the slow timescale of the cooling/quench phase that limits the period [Fig. 4]. Surface trajectories traverse the full S-curve of the slow manifold between two folds, while the bulk remains buffered on the swollen branch and traces only a small loop two orders of magnitude smaller in extent. The same diffusion barrier that organises the slow phase prevents global collapse from invading the gel interior, fixing the LCST-front penetration depth at $\xi_{\text{LCST}} \simeq 0.885$ at the working point and leaving the inner 88.5% of the slab permanently below the LCST.

A closed-form parameter inversion of the homogeneous-SS conditions exposes the limits of 0D linear stability as a predictor for the PDE. The (B_T, S_χ) plane self-folds into a Whitney cusp; the collapsed-only subregion, where no swollen homogeneous branch exists for a 0D linearisation to sit on, covers $\sim 51\%$ of the displayed plane and includes both the working point and a deep representative C-cell on which the PDE nevertheless realises the LCST-front cycle [Fig. 7(a)]. The cycle is destroyed at the upper- S_χ exit through a global saddle–node-on-invariant-circle (SNIC) bifurcation, with the period divergence following $T \propto (S_\chi^c - S_\chi)^{-1/2}$ to fitting accuracy more than twice that of the alternative homoclinic log law [Fig. 7(b)].

Linear stability of any homogeneous branch is therefore at best a necessary, never a sufficient, condition for this gel to oscillate.

Initial-condition sweeps in the bistable C-cell uncover a hidden hot-runaway steady state coexisting with the cycle in the same parameter cell at the same bath conditions [Fig. 8]. The basin of the cycle covers only $\sim 13\%$ of an 11×11 IC grid and the runaway dominates the rest, explaining why the default cold-IC sweep [Fig. 2] never sees the runaway. A 1D slice through this basin produces a sharp jump in the long-time bulk-mean $\langle J \rangle$ at $\theta_0^* \approx 2.4$ for the cold seed ($J_0 = 1.30$), and the pre-collapsed seed ($J_0 = 0.20$) terminates in the runaway for every initial thermal pulse—a falsifiable IC-induced hysteresis prediction that the spatially uniform CSTR analogue [18] does not produce.

Two predictions follow that are accessible to existing PNIPAM/exothermic-reaction platforms [1, 13, 14]. First, the SNIC scaling at the upper- S_χ exit gives a square-root critical slowing of the surface-volume oscillation period that should be observable as ΔT_∞ or the substrate concentration c_b is approached towards the regime boundary; the alternative scenario (homoclinic) is sharply discriminable by the same scan. Second, the IC-induced hysteresis at the C-cell predicts that a sufficiently warm or partially-collapsed seed of the same gel, under identical bath conditions, locks into a hot-runaway state with $\langle J \rangle$ roughly three times the cycle’s mean, and that this state is not accessible from the canonical cold IC. Both predictions are ergodicity-breaking signatures of the spatially generated slow manifold and would not arise in a well-mixed analogue.

Several questions remain that the present work brackets but does not settle. The slab geometry is the simplest setting in which the spatially generated slow manifold is realised; how curvature (spherical/membrane) modifies the basin of the cycle, the runaway, and the SNIC scaling is the subject of an ongoing study. The parameter-space optimisation of interior reactant penetration—an application-side question for autonomous chemical-delivery contexts—also lies beyond the present mechanism-focused scope. We expect both questions to inherit the necessary-but-not-sufficient character of the 0D linear analysis demonstrated here.

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[Acknowledgments placeholder.]

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Appendix A: Model Construction

We formulate a one-dimensional Lagrangian model for the coupled thermo-chemo-poroelastic dynamics of a thermoresponsive gel slab undergoing an exothermic reaction. The primary variables are the local swelling ratio $J(X, t)$, reactant concentration $c(X, t)$, and temperature $T(X, t)$.

1. Kinematics

Consider a gel slab of reference (as-prepared) half-thickness H_0 . We adopt a material (Lagrangian) coordinate $X \in [0, H_0]$, with $X = 0$ denoting the symmetry plane and $X = H_0$ the free surface. The current position of a material point is $x(X, t)$, and the local volume ratio is

$$J(X, t) = \frac{\partial x}{\partial X} > 0. \quad (\text{A1})$$

In one dimension, J is the local swelling ratio. The polymer volume fraction is a dependent variable determined by mass conservation of the network:

$$\phi(X, t) = \frac{\phi_{p0}}{J(X, t)}, \quad (\text{A2})$$

where ϕ_{p0} is the polymer volume fraction in the reference (as-prepared) state; $\phi_{p0} = 0.15$ corresponds to a gel that is 85% solvent by volume at preparation. The current thickness of the slab is

$$h(t) = x(H_0, t) = \int_0^{H_0} J(X, t) dX. \quad (\text{A3})$$

2. Governing equations

a. Swelling-permeation equation

The evolution of J is governed by solvent conservation:

$$\frac{\partial J}{\partial t} = -\frac{\partial Q_s}{\partial X}, \quad (\text{A4})$$

where Q_s is the solvent flux relative to the polymer network, driven by chemical-potential gradients:

$$Q_s = -M_s(J, T) \frac{\partial \mu_s}{\partial X}. \quad (\text{A5})$$

Here $M_s(J, T)$ is the permeation mobility and μ_s is the solvent chemical potential.

b. Reactant transport equation

The reactant concentration $c(X, t)$ (moles per unit current total volume) evolves according to

$$\frac{\partial(Jc)}{\partial t} + \frac{\partial N_c}{\partial X} = -J R(c, T, J), \quad (\text{A6})$$

with the total reactant flux

$$N_c = c Q_s - D_c(J, T) \frac{\partial c}{\partial X}. \quad (\text{A7})$$

The first term represents advective transport by the solvent flow, and the second term Fickian diffusion within the pore fluid. The effective diffusivity $D_c(J, T)$ depends on the local microstructure via J .

c. Heat equation

The temperature field satisfies

$$C_{\text{eff}}(J) \frac{\partial T}{\partial t} = \frac{\partial}{\partial X} \left(K_T(J) \frac{\partial T}{\partial X} \right) - \frac{\partial}{\partial X} (c_p^\ell T Q_s) + (-\Delta H_r) J R(c, T, J), \quad (\text{A8})$$

where $C_{\text{eff}}(J)$ is the effective volumetric heat capacity, $K_T(J)$ the effective thermal conductivity, c_p^ℓ the solvent heat capacity per unit volume, and $-\Delta H_r > 0$ the exothermic reaction enthalpy. The three right-hand-side terms represent thermal conduction, enthalpy advection by the solvent flow, and reaction heat generation, respectively.

3. Free energy and chemical potential

The Helmholtz free-energy density per unit reference volume is written as

$$\Psi(J, T) = \Psi_{\text{el}}(J) + \Psi_{\text{mix}}(\phi, T) + \frac{\kappa_J}{2} \left(\frac{\partial J}{\partial X} \right)^2, \quad (\text{A9})$$

where $\phi = \phi_{p0}/J$, and the three contributions are the elastic energy of the network, the Flory–Huggins mixing energy, and a gradient-energy regularization.

For the polymer network we adopt a neo-Hookean elastic energy appropriate for uniaxial swelling in the slab geometry:

$$\Psi_{\text{el}}(J) = \frac{G}{2}(J^2 - 1 - 2 \ln J), \quad (\text{A10})$$

where G is the shear modulus of the reference-state network.

The mixing free-energy density takes the standard Flory–Huggins form [10]:

$$\Psi_{\text{mix}}(\phi, T) = \frac{R_g T}{v_s} \left[(1 - \phi) \ln(1 - \phi) + \chi(T, \phi) \phi(1 - \phi) \right], \quad (\text{A11})$$

with R_g the gas constant, v_s the solvent molar volume, and the Flory–Huggins interaction parameter

$$\chi(T, \phi) = \chi_0(T) + \chi_1 \phi. \quad (\text{A12})$$

The temperature-dependent part $\chi_0(T)$ encodes the LCST behavior of the gel; χ_1 captures the leading concentration dependence.

The solvent chemical potential is obtained as $\mu_s = v_s \partial \Psi / \partial (1 - \phi) \cdot \partial (1 - \phi) / \partial J$ [12]. Evaluating this derivative yields the mixing contribution

$$\frac{\mu_{\text{mix}}}{R_g T / v_s} = \ln(1 - \phi) + \phi + \chi(T, \phi) \phi^2, \quad (\text{A13})$$

and the elastic contribution

$$\frac{\mu_{\text{el}}}{R_g T / v_s} = \frac{G v_s}{R_g T} (J - J^{-1}). \quad (\text{A14})$$

The full chemical potential, including the gradient-energy regularization, is

$$\mu_s = \mu_{\text{mix}}(J, T) + \mu_{\text{el}}(J) - \kappa_J \frac{\partial^2 J}{\partial X^2}. \quad (\text{A15})$$

4. Reaction kinetics

The volumetric reaction rate takes the Arrhenius form modulated by a collapse-dependent accessibility factor:

$$R(c, T, J) = k_0 c^n \exp\left(-\frac{E_a}{R_g T}\right) a(J), \quad (\text{A16})$$

where k_0 is the pre-exponential factor, n the reaction order, and E_a the activation energy.

The activity factor $a(J)$ models the suppression of reaction due to reduced free solvent volume upon collapse:

$$a(J) = \left(1 - \frac{\phi_{p0}}{J}\right)^{m_{\text{act}}} = (1 - \phi)^{m_{\text{act}}}. \quad (\text{A17})$$

When the gel collapses ($J \rightarrow \phi_{p0}$, $\phi \rightarrow 1$), the activity vanishes, shutting off the reaction.

The effective transport coefficients similarly depend on the microstructure:

$$D_c(J) = D_0 \left(1 - \frac{\phi_{p0}}{J}\right)^{m_{\text{diff}}}, \quad (\text{A18})$$

$$M_s(J) = M_0 \left(1 - \frac{\phi_{p0}}{J}\right)^{m_{\text{mob}}}, \quad (\text{A19})$$

so that both solute diffusion and solvent permeation are suppressed in the collapsed state.

5. Boundary conditions

At the symmetry plane $X = 0$:

$$Q_s = 0, \quad N_c = 0, \quad \frac{\partial T}{\partial X} = 0. \quad (\text{A20})$$

At the free surface $X = H_0$:

a. Solvent exchange. A Robin-type condition couples the surface chemical potential to the bath:

$$Q_s = L_\mu (\mu_s - \mu_{\text{bath}}), \quad (\text{A21})$$

where L_μ is a surface transfer coefficient. In the limit of rapid equilibration, $\mu_s = \mu_{\text{bath}}$.

b. Reactant supply.

$$N_c = k_c (c - c_b), \quad (\text{A22})$$

where c_b is the constant bath concentration.

c. Heat exchange.

$$-K_T \frac{\partial T}{\partial X} = h_T (T - T_\infty), \quad (\text{A23})$$

where h_T is the surface heat-transfer coefficient and T_∞ the ambient temperature.

d. Mechanical boundary. The free surface is traction-free. In the J -formulation this is implicitly satisfied by the chemical-potential balance.

Appendix B: Nondimensionalization

1. Scaling

We introduce the following dimensionless variables. Length is scaled by the reference thickness:

$$X = H_0 \tilde{x}, \quad \tilde{x} \in [0, 1]. \quad (\text{B1})$$

The swelling ratio J is already dimensionless. Reactant concentration is normalized by the bath value:

$$c = c_b u. \quad (\text{B2})$$

Temperature is measured relative to ambient, scaled by the adiabatic reaction temperature rise:

$$T = T_\infty + \Delta T_* \theta, \quad \Delta T_* = \frac{(-\Delta H_r) c_b}{C_*}, \quad (\text{B3})$$

where C_* is a representative volumetric heat capacity.

The chemical potential is scaled by the thermal mixing energy:

$$\mu_s = \mu_* m, \quad \mu_* = \frac{R_g T_\infty}{v_s}. \quad (\text{B4})$$

Time is scaled by the swelling/permeation time:

$$\tau_s = \frac{H_0^2}{M_0 \mu_*}, \quad (\text{B5})$$

where M_0 is a reference mobility. The solvent and reactant flux scales follow:

$$Q_s = \frac{H_0}{\tau_s} q, \quad N_c = \frac{c_b H_0}{\tau_s} n. \quad (\text{B6})$$

Material functions are factored as reference value times dimensionless function:

$$M_s = M_0 \mathcal{M}(J, \theta), \quad D_c = D_0 \mathcal{D}(J, \theta), \quad (\text{B7})$$

$$C_{\text{eff}} = C_* \mathcal{C}(J), \quad K_T = K_* \mathcal{K}(J). \quad (\text{B8})$$

2. Dimensionless reaction rate

Defining the reference reaction rate

$$R_0 = k_0 c_b^n \exp\left(-\frac{E_a}{R_g T_\infty}\right), \quad (\text{B9})$$

we write $R = R_0 \hat{R}(u, \theta, J)$ with

$$\hat{R}(u, \theta, J) = u^n a(J) \exp\left[\frac{\Gamma_A \theta}{1 + \varepsilon_T \theta}\right], \quad (\text{B10})$$

where

$$\varepsilon_T = \frac{\Delta T_*}{T_\infty}, \quad \Gamma_A = \frac{E_a \Delta T_*}{R_g T_\infty^2}. \quad (\text{B11})$$

When $\varepsilon_T \ll 1$, this simplifies to $\hat{R} \approx u^n a(J) e^{\Gamma_A \theta}$.

3. Dimensionless chemical potential

The dimensionless chemical potential is

$$m = m_{\text{mix}}(J, \theta) + m_{\text{el}}(J) - \ell^2 \frac{\partial^2 J}{\partial \tilde{x}^2}, \quad (\text{B12})$$

with the interface parameter

$$\ell^2 = \frac{\kappa_J}{\mu_* H_0^2}. \quad (\text{B13})$$

The mixing part retains the Flory–Huggins form (neglecting the $T/T_\infty \approx 1 + \varepsilon_T \theta \approx 1$ prefactor, valid when $\varepsilon_T \ll 1$):

$$m_{\text{mix}} = \ln(1 - \phi) + \phi + \chi(\theta, \phi) \phi^2, \quad \phi = \frac{\phi_{p0}}{J}, \quad (\text{B14})$$

with the dimensionless interaction parameter

$$\chi(\theta, \phi) = \chi_\infty + S_\chi \theta + \chi_1 \phi, \quad (\text{B15})$$

where $\chi_\infty = \chi_0(T_\infty)$ and the thermo-swelling sensitivity is

$$S_\chi = \Delta T_* \left. \frac{d\chi_0}{dT} \right|_{T_\infty}. \quad (\text{B16})$$

The elastic contribution follows from Eq. (A14):

$$m_{\text{el}} = \Omega_e (J - J^{-1}), \quad \Omega_e = \frac{G v_s}{R_g T_\infty}, \quad (\text{B17})$$

measuring the network stiffness relative to the thermal mixing energy.

4. Dimensionless governing equations

Dropping tildes for clarity, the dimensionless system reads:

a. Swelling equation.

$$\frac{\partial J}{\partial t} = \frac{\partial}{\partial x} \left(\mathcal{M}(J, \theta) \frac{\partial m}{\partial x} \right), \quad (\text{B18})$$

b. Chemical potential.

$$m = m_{\text{mix}}(J, \theta) + m_{\text{el}}(J) - \ell^2 \frac{\partial^2 J}{\partial x^2}, \quad (\text{B19})$$

c. Reactant equation.

$$\frac{\partial(Ju)}{\partial t} + \frac{\partial}{\partial x} \left(u q - \delta \mathcal{D}(J, \theta) \frac{\partial u}{\partial x} \right) = -\text{Da} J \hat{R}(u, \theta, J), \quad (\text{B20})$$

where $q = -\mathcal{M}(J, \theta) \partial_x m$ and

$$\delta = \frac{D_0}{M_0 \mu_*} \quad (\text{B21})$$

is the ratio of solute diffusion speed to solvent permeation speed.

The Damköhler number, measuring reaction rate relative to swelling rate, is

$$\text{Da} = \tau_s k_0 c_b^{n-1} \exp\left(-\frac{E_a}{R_g T_\infty}\right). \quad (\text{B22})$$

d. Heat equation.

$$\begin{aligned} \mathcal{C}(J) \frac{\partial \theta}{\partial t} = & \alpha \frac{\partial}{\partial x} \left(\mathcal{K}(J) \frac{\partial \theta}{\partial x} \right) + \text{Da} J \hat{R}(u, \theta, J) \\ & - \alpha \text{Pe}_T \frac{\partial}{\partial x} [(\Theta_\infty + \theta) q], \end{aligned} \quad (\text{B23})$$

with

$$\alpha = \frac{K_* \tau_s}{C_* H_0^2} = \frac{\tau_s}{\tau_T}, \quad \text{Pe}_T = \frac{c_p^\ell M_0 \mu_*}{K_*}, \quad \Theta_\infty = \frac{T_\infty}{\Delta T_*}, \quad (\text{B24})$$

where $\tau_T = C_* H_0^2 / K_*$ is the thermal diffusion time. When $\text{Pe}_T \ll 1$, the convective term may be dropped.

e. *Constraint.*

$$\phi = \frac{\phi_{p0}}{J}. \quad (\text{B25})$$

5. Dimensionless boundary conditions

At $x = 0$ (symmetry):

$$q = 0, \quad u q - \delta \mathcal{D} u_x = 0, \quad \theta_x = 0. \quad (\text{B26})$$

At $x = 1$ (free surface):

$$q = \text{Bi}_\mu (m - m_b), \quad (\text{B27})$$

$$u q - \delta \mathcal{D} u_x = \text{Bi}_c (u - 1), \quad (\text{B28})$$

$$-\alpha \mathcal{K}(J) \theta_x = \text{Bi}_T \theta, \quad (\text{B29})$$

where the surface Biot numbers for permeation and mass transfer are

$$\text{Bi}_\mu = \frac{L_\mu H_0}{M_0}, \quad \text{Bi}_c = \frac{k_c H_0}{M_0 \mu_*}, \quad (\text{B30})$$

and the thermal exchange number is

$$\text{Bi}_T = \frac{h_T \tau_s}{C_* H_0} = \alpha \frac{h_T H_0}{K_*}, \quad (\text{B31})$$

which measures the surface heat loss rate directly on the swelling timescale τ_s . The standard thermal Biot number $h_T H_0 / K_* = \text{Bi}_T / \alpha$ is recovered by factoring out α .

6. Controlling parameter groups

The oscillatory dynamics are governed by the following dimensionless groups:

- Da — Damköhler number: strength of reaction heat source relative to swelling.
- $\alpha = \tau_s / \tau_T$ — thermal diffusion relative to swelling.
- S_χ — thermo-swelling sensitivity: how strongly temperature shifts the interaction parameter.
- Γ_A — Arrhenius feedback strength.
- δ — solute diffusion to permeation ratio.
- ℓ — interface regularization length.
- Bi_c — surface reactant Biot number.
- Bi_T — thermal exchange number (surface heat loss on the swelling timescale).

Together with the working-point parameters ϕ_{p0} and χ_∞ , these form the minimal set controlling the onset of self-oscillation. In physical terms, oscillation requires that the fast positive feedback (reaction $\rightarrow$ heating $\rightarrow$ collapse) and the slow negative feedback (collapse $\rightarrow$ transport suppression $\rightarrow$ cooling $\rightarrow$ re-swelling) operate on well-separated time scales.

Appendix C: Dispersion Relation Derivation

We derive the dispersion matrix $\mathbf{A}(k)$ introduced in Eq. (5). The $k = 0$ matrix $\mathbf{A}_0$ is obtained by volume-averaging the governing equations over the slab domain and incorporating the Robin boundary fluxes as lumped source/sink terms. The k^2 and k^4 matrices $\mathbf{D}_2$ and $\mathbf{D}_4$ collect the bulk diffusion contributions from a point-wise normal-mode analysis; cross-coupling through advection ($u_x q$ and $\theta_x q$) vanishes identically at the uniform base state because both $q_0 = 0$ and the field gradients vanish.

Starting from the mixed formulation (D1)–(D4), the chemical potential entering the swelling flux is

$$m = f(J, \theta) - \ell^2 J_{\xi\xi}, \quad (\text{C1})$$

with $f(J, \theta) = m_{\text{mix}}(J, \theta) + m_{\text{el}}(J)$. We perturb about the uniform base state with normal modes $(\hat{J}, \hat{u}, \hat{\theta})^T e^{\sigma t + i k \xi}$ and define the partial derivatives

$$f_J = \left. \frac{\partial f}{\partial J} \right|_0, \quad f_\theta = \left. \frac{\partial f}{\partial \theta} \right|_0, \quad (\text{C2a})$$

$$R_J = \left. \frac{\partial \hat{R}}{\partial J} \right|_0, \quad R_u = \left. \frac{\partial \hat{R}}{\partial u} \right|_0, \quad (\text{C2b})$$

$$R_\theta = \left. \frac{\partial \hat{R}}{\partial \theta} \right|_0, \quad (\text{C2c})$$

and the base-state transport coefficients

$$\mathcal{M}_0 = \mathcal{M}(J_0, \theta_0), \quad \mathcal{D}_0 = \delta \mathcal{D}(J_0, \theta_0), \quad (\text{C3a})$$

$$\mathcal{K}_0 = \alpha \mathcal{K}(J_0), \quad \mathcal{C}_0 = \mathcal{C}(J_0). \quad (\text{C3b})$$

The $k = 0$ matrix $\mathbf{A}_0$ contains the reaction and boundary-exchange terms:

$$\mathbf{A}_0 = \begin{bmatrix} a_{11} & 0 & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad (\text{C4})$$

with entries

$$a_{11} = -\text{Bi}_\mu f_J, \quad a_{13} = -\text{Bi}_\mu f_\theta, \quad (\text{C5a})$$

$$a_{21} = \frac{u_0 \text{Bi}_\mu f_J - \text{Da}(\hat{R}_0 + J_0 R_J)}{J_0}, \quad (\text{C5b})$$

$$a_{22} = \frac{-\text{Bi}_c - \text{Da} J_0 R_u}{J_0}, \quad (\text{C5c})$$

$$a_{23} = \frac{u_0 \text{Bi}_\mu f_\theta - \text{Da} J_0 R_\theta}{J_0}, \quad (\text{C5d})$$

$$a_{31} = \frac{\text{Da}(\hat{R}_0 + J_0 R_J)}{\mathcal{C}_0}, \quad a_{32} = \frac{\text{Da} J_0 R_u}{\mathcal{C}_0}, \quad (\text{C5e})$$

$$a_{33} = \frac{-\text{Bi}_T + \text{Da} J_0 R_\theta}{\mathcal{C}_0}. \quad (\text{C5f})$$

Note that $a_{12} = 0$ because the swelling equation does not depend directly on the reactant concentration.

The k^2 contribution collects diffusive transport:

$$\mathbf{D}_2 = \begin{bmatrix} -\mathcal{M}_0 f_J & 0 & -\mathcal{M}_0 f_\theta \\ 0 & -\mathcal{D}_0/J_0 & 0 \\ 0 & 0 & -\mathcal{K}_0/\mathcal{C}_0 \end{bmatrix}. \quad (\text{C6})$$

The $(1, 1)$ entry $-\mathcal{M}_0 f_J$ changes sign when $f_J < 0$ (spinodal region), acting as a negative diffusivity for J .

The only k^4 contribution arises from the Cahn–Hilliard regularization:

$$\mathbf{D}_4 = \begin{bmatrix} -\mathcal{M}_0 \ell^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (\text{C7})$$

Together, $\mathbf{A}(k) = \mathbf{A}_0 + k^2 \mathbf{D}_2 + k^4 \mathbf{D}_4$ yields a 3×3 eigenvalue problem for each wavenumber k .

Appendix D: Numerical Method

We describe a cell-centered finite-volume spatial discretization for the dimensionless system derived in Appendix B. Time integration uses the implicit BDF method provided by `SCIPY.INTEGRATE.SOLVE_IVP`: the spatial operators below define the right-hand side of the resulting ODE system (method of lines). In the notation below, superscript n denotes the current time level and $n+1$ the new level; starred quantities ($\mathcal{M}_i^*$, θ_i^* , $\mathcal{C}_i^*$) are evaluated at the most recent available state, and $\sharp$ -quantities ($J_i^\sharp$, $u_i^\sharp$) at the values from the preceding fractional stage when operator splitting is used.

1. Mixed formulation

To avoid discretizing fourth-order derivatives directly, we introduce the auxiliary variables m , q , n , and h and write the system as

$$J_t + q_x = 0, \quad q = -\mathcal{M}(J, \theta) m_x, \quad (\text{D1})$$

$$m = f(J, \theta) - \ell^2 J_{xx}, \quad (\text{D2})$$

$$(Ju)_t + n_x = -\text{Da } J \hat{R}, \quad n = u q - \delta \mathcal{D}(J, \theta) u_x, \quad (\text{D3})$$

$$\mathcal{C}(J) \theta_t + h_x = \text{Da } J \hat{R}, \quad h = -\alpha \mathcal{K}(J) \theta_x, \quad (\text{D4})$$

where $f(J, \theta) = m_{\text{mix}}(J, \theta) + m_{\text{el}}(J)$. This mixed form requires only C^0 regularity (no second-derivative continuity) and naturally accommodates Robin boundary conditions.

2. Spatial discretization

We use a uniform mesh of N cells:

$$\Delta x = \frac{1}{N}, \quad x_i = \left(i - \frac{1}{2}\right) \Delta x, \quad i = 1, \dots, N. \quad (\text{D5})$$

Face positions are $x_{i+1/2} = i \Delta x$ for $i = 0, \dots, N$. Cell-centered unknowns are J_i , m_i , u_i , θ_i ; face fluxes are $q_{i+1/2}$, $n_{i+1/2}$, $h_{i+1/2}$.

a. Swelling equation

The conservative update reads

$$\frac{J_i^{n+1} - J_i^n}{\Delta t} + \frac{q_{i+1/2}^{n+1} - q_{i-1/2}^{n+1}}{\Delta x} = 0, \quad (\text{D6})$$

with the face flux

$$q_{i+1/2}^{n+1} = -\mathcal{M}_{i+1/2}^* \frac{m_{i+1}^{n+1} - m_i^{n+1}}{\Delta x}. \quad (\text{D7})$$

The face mobility $\mathcal{M}_{i+1/2}^*$ is evaluated using the harmonic mean

$$\mathcal{M}_{i+1/2}^* = \frac{2 \mathcal{M}_i^* \mathcal{M}_{i+1}^*}{\mathcal{M}_i^* + \mathcal{M}_{i+1}^*}, \quad (\text{D8})$$

which is more robust than arithmetic averaging for degenerate mobilities.

b. Chemical potential

The discrete chemical potential is

$$m_i^{n+1} = f(J_i^{n+1}, \theta_i^*) - \ell^2 \frac{J_{i+1}^{n+1} - 2J_i^{n+1} + J_{i-1}^{n+1}}{\Delta x^2}, \quad (\text{D9})$$

where the natural boundary condition $\partial_x J = 0$ is imposed via ghost cells:

$$J_0^{n+1} = J_1^{n+1}, \quad J_{N+1}^{n+1} = J_N^{n+1}. \quad (\text{D10})$$

c. Reactant equation

To preserve conservation, we update the product $W_i = J_i u_i$:

$$\frac{W_i^{n+1} - W_i^n}{\Delta t} + \frac{n_{i+1/2}^{n+1} - n_{i-1/2}^{n+1}}{\Delta x} = -\text{Da } J_i^\sharp \hat{R}(u_i^\sharp, \theta_i^\sharp, J_i^\sharp), \quad (\text{D11})$$

followed by reconstruction $u_i^{n+1} = W_i^{n+1}/J_i^{n+1}$.

The reactant face flux combines upwind advection and central diffusion:

$$n_{i+1/2}^{n+1} = q_{i+1/2}^{n+1} u_{i+1/2}^{\text{up}} - \delta \mathcal{D}_{i+1/2}^* \frac{u_{i+1}^{n+1} - u_i^{n+1}}{\Delta x}, \quad (\text{D12})$$

where the upwind value is

$$u_{i+1/2}^{\text{up}} = \begin{cases} u_i^{n+1}, & q_{i+1/2}^{n+1} \geq 0, \\ u_{i+1}^{n+1}, & q_{i+1/2}^{n+1} < 0. \end{cases} \quad (\text{D13})$$

Upwinding is essential to suppress spurious oscillations when advection dominates over diffusion.

d. Heat equation

The thermal update is

$$C_i^* \frac{\theta_i^{n+1} - \theta_i^n}{\Delta t} + \frac{h_{i+1/2}^{n+1} - h_{i-1/2}^{n+1}}{\Delta x} = \text{Da } J_i^\# \widehat{R}(u_i^\#, \theta_i^\#, J_i^\#), \quad (\text{D14})$$

with the heat flux (neglecting enthalpy advection)

$$h_{i+1/2}^{n+1} = -\alpha \mathcal{K}_{i+1/2}^* \frac{\theta_{i+1}^{n+1} - \theta_i^{n+1}}{\Delta x}. \quad (\text{D15})$$

3. Boundary conditions

At the left boundary ($x = 0$):

$$q_{1/2}^{n+1} = 0, \quad n_{1/2}^{n+1} = 0, \quad h_{1/2}^{n+1} = 0. \quad (\text{D16})$$

At the right boundary ($x = 1$), the Robin conditions become face fluxes:

$$q_{N+1/2}^{n+1} = \text{Bi}_\mu (m_N^{n+1} - m_b), \quad (\text{D17})$$

$$n_{N+1/2}^{n+1} = \text{Bi}_c (u_N^{n+1} - 1), \quad (\text{D18})$$

$$h_{N+1/2}^{n+1} = \text{Bi}_T \theta_N^{n+1}. \quad (\text{D19})$$

4. Time integration

The primary integration strategy is the method of lines (MOL): the spatial discretization above defines a system of $4N$ coupled ODEs that is advanced in time by the variable-order implicit BDF solver (SCIPY.INTEGRATE.SOLVE_IVP with `method='BDF'` and a precomputed Jacobian sparsity pattern). Adaptive time stepping and error control are handled by the library.

For parameter regimes where the BDF solver encounters difficulties, an alternative IMEX operator-splitting strategy is available, in which each time step proceeds sequentially:

a. Stage 1: Swelling (J, m, q). Given (J^n, u^n, θ^n) , solve the coupled nonlinear system (D6)–(D9) for $(J^{n+1}, m^{n+1}, q^{n+1})$ using Newton iteration. The mobility and thermodynamic function f are evaluated at the current iterate $(J^{(k)}, \theta^{(k)})$.

b. Stage 2: Reactant (W, u). With J^{n+1} and q^{n+1} known, solve the linear (or mildly nonlinear) system (D11)–(D12) for W^{n+1} , then recover $u^{n+1} = W^{n+1}/J^{n+1}$. Diffusion is treated implicitly; the reaction source may be linearized about the current temperature.

c. Stage 3: Temperature (θ). With J^{n+1} and u^{n+1} known, solve the implicit heat equation (D14)–(D15) for θ^{n+1} . If the Arrhenius source term is stiff, a Newton linearization of the source with respect to θ is applied.

5. Fully coupled Newton formulation

For problems requiring tight coupling, the unknowns are assembled into a single vector

$$\mathbf{Y} = (J_1, \dots, J_N, m_1, \dots, m_N, W_1, \dots, W_N, \theta_1, \dots, \theta_N)^T \quad (\text{D20})$$

and the residual vector $\mathbf{R}(\mathbf{Y})$ collects all discrete equations (D6)–(D15) with boundary conditions (D16)–(D19). The system is solved by Newton's method:

$$\mathbf{J}_{\text{ac}} \delta \mathbf{Y} = -\mathbf{R}(\mathbf{Y}^{(k)}), \quad \mathbf{Y}^{(k+1)} = \mathbf{Y}^{(k)} + \delta \mathbf{Y}, \quad (\text{D21})$$

where $\mathbf{J}_{\text{ac}}$ is the Jacobian matrix. Because all couplings are local in one dimension, the Jacobian is sparse and banded; its sparsity pattern is precomputed and supplied to the BDF integrator for efficient graph-coloring-based finite-difference Jacobian evaluation.

6. Implementation notes

1. The conserved variable $W = Ju$ must be used for the reactant update, not a non-conservative discretization of u alone. This ensures that boundary supply and internal reaction remain in balance even during rapid swelling or collapse.
2. Upwind discretization of the advective flux uq is mandatory, not optional. Central differencing produces non-physical oscillations whenever $|q| \Delta x / (\delta \mathcal{D}) \gg 1$.
3. The fourth-order swelling–chemical-potential system must be split into two second-order equations (mixed form). Direct discretization of J_{xxx} leads to wider stencils and difficulties with natural boundary conditions.

Appendix E: Spinodal Decomposition Verification

The finite-wavenumber instability identified in Sec. III corresponds to spinodal decomposition of the gel within the Flory–Huggins two-phase region. To verify that this mode is a genuine feature of the model—not merely an

artifact of the linearization—we perform a direct numerical simulation of the pure Cahn–Hilliard dynamics: the reaction is switched off ($\text{Da} = 0$), the temperature is held fixed at $\theta = 1.5$ (above the LCST), and no-flux boundary conditions are imposed at both ends.

The simulation starts from a spatially uniform collapsed state $J_0 = 0.30$ ($\phi = 0.50$) with small random perturbations ($\delta J/J_0 \sim 0.1\%$), using $N = 301$ grid points and $\ell = 0.01$. At this state, $f_J = -1.76 < 0$, confirming that the uniform state is inside the spinodal. The predicted fastest-growing wavenumber is $k^* = \sqrt{-f_J/(2\ell^2)} \approx 94$, corresponding to a domain spacing $\lambda^* \approx 0.067 H_0$. The maximum growth rate is $\sigma^* = \mathcal{M}_0 f_J^2/(4\ell^2) \approx 3850$, where $\mathcal{M}_0 = (1 - \phi)^{m_{\text{mob}}} = 0.50$ is the base-state mobility.

Figure 9 shows the result. The initially uniform state spontaneously separates into alternating swollen ($J \approx 0.7$) and collapsed ($J \approx 0.2$) domains within $\tau \lesssim 0.1$

[Fig. 9(a,b)]. The Fourier amplitude at $k \approx k^*$ [Fig. 9(c)] grows exponentially at the predicted rate $2\sigma^*$ over five orders of magnitude before saturating as the system enters the nonlinear regime. The full Fourier spectrum [Fig. 9(d)] confirms that the dominant wavenumber during the linear growth phase ($\tau \lesssim 0.003$) peaks near the predicted k^* ; at later times, the spectrum shifts to lower k as domains coarsen through coalescence.

This verification establishes two points: (i) the model faithfully captures Cahn–Hilliard spinodal physics with quantitative agreement between the linear prediction and the nonlinear simulation; (ii) the spinodal mode operates on a timescale ($\sigma^{*-1} \sim 3 \times 10^{-4} \tau$) that is three orders of magnitude faster than the Hopf oscillation period ($T \sim 20 \tau$), explaining why it does not interfere with the self-sustained oscillation dynamics when the reaction is active.

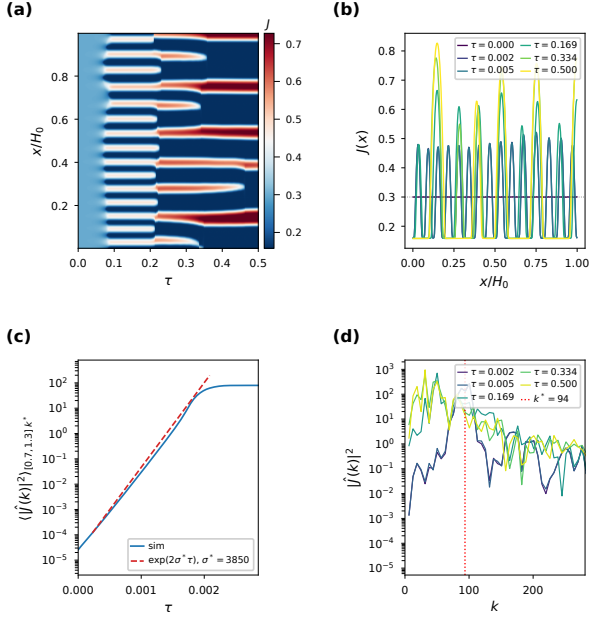


FIG. 9. Spinodal decomposition in the gel model with $Da = 0$ (no reaction), $\theta = 1.5$ (fixed), $J_0 = 0.30$, $\ell = 0.01$, $N = 301$, no-flux boundaries. **(a)** Kymograph of $J(x, \tau)$ showing spontaneous domain formation and subsequent coarsening. **(b)** Spatial profiles at selected times. **(c)** Fourier amplitude $|\hat{J}(k^*)|^2$ averaged over $k \in [0.7k^*, 1.3k^*]$; the predicted exponential growth at rate $2\sigma^*$ (red dashed) matches the simulation over five decades before nonlinear saturation. **(d)** Fourier power spectrum $|\hat{J}(k)|^2$; the earliest curves peak near $k^* \approx 94$ (red dotted), shifting to lower k during coarsening.